8 Antennas

INTRODUCTION

- Antenna is a device that receives and transmits electromagnetic waves.
- Electromagnetic waves are generally generated by the sources, which have time-varying changes and currents.
  An antenna is also known as a radiator or an aerial.
- It also behaves as a coupler between a guided structure such as a waveguide or transmission line and space or vice versa.
- It can be said that a antenna is a transducer that mainly converts voltage or current signals into EM waves or vice-versa.

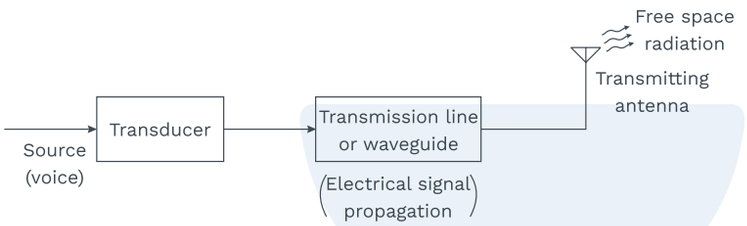


Figure 8.1: Transmission

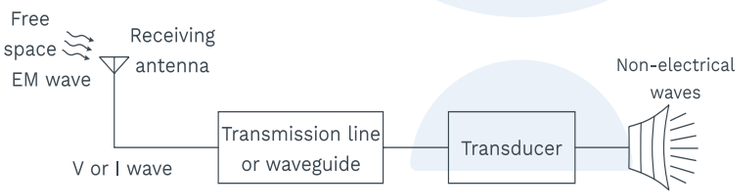


Figure 8.2: Reception

- The basic equation of radiation is given by-

$$L\frac{di}{dt} = Q\frac{dv}{dt}$$

This equation shows that change in the current flowing in a length (L) results in the acceleration of radiating charge.

where L = Length of antenna

$\frac{di}{dt}$ = Changing current

$\frac{dv}{dt}$ = Accelerated charge

- If a charge is not moving, then the current must be zero, and hence no radiation will be there.

Rack your Brain

Why can't we use multiple reflectors?

And, when a charge is moving with an uniform velocity, there will be no radiation if the wire is straight or infinitely extended.

- If a charge is moving with a uniform velocity and radiation will occur if the wire is curved, bent, discontinuous, or terminated.
  And if the charge is oscillating in time-motion, it must radiates even if the wire is straight.
- To determine the radiation field regardless of the type of antenna, the following steps can be employed-
  (i) At first, choose an appropriate coordinate system and then evaluate the magnetic vector potential $(\vec{A})$.
  (ii) Determine $\vec{H}$ (magnetic field intensity) by $\vec{B} = \mu\vec{H} = \nabla \times \vec{A}$ formula.
  (iii) Calculate electric field $(\vec{E})$ using $\nabla \times \vec{H} = j\omega \in (\vec{E})$, considering a lossless medium ($\sigma = 0$).
  (iv) Find the field and evaluate the time-average power radiated by using the formula-

$$P_{rad} = \int_s \vec{P}_{avg} \cdot d\vec{S}$$

where, $\vec{P}_{avg} = \frac{1}{2} Re[\vec{E} \times \vec{H}^*]$

- While following these points step by step and analyzing the antenna radiation field, we generally-
  Consider the spherical coordinate system as we know that the antenna radiates in all directions primarily. Thus, we assume the spherical radiation symmetry.

Previous Year's Questions

The electric field $\vec{E}$ and the magnetic field $\vec{H}$ of a short dipole antenna satisfies the condition-

(A) The r component of $\vec{E}$ is equal to zero.
(B) Both r and θ components of $\vec{E}$ equal to zero.
(C) The θ component of $\vec{E}$ dominates the r component in the far-field region.
(D) The θ and ϕ component of $\vec{H}$ are of the same order of magnitude in the near-field region.

- We usually use Maxwell's equations to derive/calculate for free space or lossless medium.
- And when it is said that the EM wave is transmitted then it mainly refers to power transmitting or power radiating.
- As we know that the direction in which a wave travels is the direction of power flow. Hence, the power radiated is calculated by the given above formula.

HERTZIAN DIPOLE

- The smallest antenna (where the length of the conductor is very small) or can say antenna having infinitesimal small current element Idl.

  i.e. $Idl \leq \frac{\lambda}{10}$

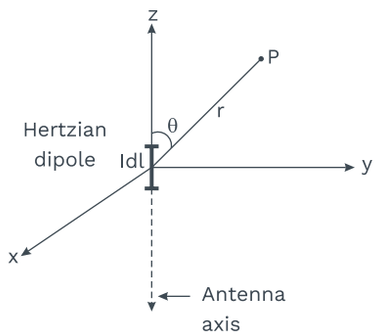


Figure 8.3: Hertzian Dipole

- Consider Idl infinitesimal conductor at the origin.
  The axis along which the antenna element is placed is considered the antenna axis.
  Here, the antenna axis is the z-axis.
- An antenna having its length less than or equal to the $\frac{1}{10^{th}}$ times of the signal's wavelength.

Rack your Brain

What is the condition of radiation by the dipole antenna?

(A) A charge must be stationary.
(B) A charge must be moving with an uniform velocity and the wire should be straight or infinite.
(C) A charge must move with a uniform velocity and the wire should be curved.
(D) The charge must oscillate in time motion.

Previous Year's Questions

Two isotropic antennas are separated by a distance of two wavelengths. If both the antennas are fed with currents of equal phase and magnitude, the number of lobes in the radiation in the horizontal plane are-

(A) 2 (B) 4
(C) 6 (D) 8

i.e., $dl \leq \frac{\lambda}{10}$

Then, that antenna will be considered as the Hertzian dipole.

- The magnetic vector potential can be expressed as-

$$\bar{A}_z = \frac{\mu[I]dl}{4\pi r}\hat{a}_z$$

where [I] is the retarded current and dl is the length of the current-carrying conductor/element.

- Let a point 'P' which is at 'r' distance from the antenna element.

  The line joining 'P' and the origin where the antenna element is placed makes 'θ' angle with the z-axis.

- 'P' is the point considered here where the potential radiation is to be calculated due to the antenna element at the origin.

- The antenna element is required to have excitation or in other words, the conductor will radiate only when it has oscillating current or oscillating charges, then only the time-varying current can be obtained.

  In this way, a time-varying current is obtained in the Idl antenna element.

- And the radiation is assumed to be calculated at 'P' also it will take some time to reach point 'P'.

  Thus, we will get the radiation after some time delay, and the current 'I' is denoted as [I] and named as retarded current.

- The retarded current [I] is the current that takes some time to reach point 'P'.

Rack your Brain

An antenna a transducer? (Yes/No)

Previous Year's Questions

Two dissimilar antennas having equal maximum directivities, then-

(A) Must have their beamwidths also equal.
(B) Cannot have their beam widths equal because they are dissimilar antennas.
(C) May not necessarily have their maximum power gains equally.
(D) Must have their effective aperture areas (capture areas) also equal.

- Hence, the current distribution at point 'P' will be evaluated after some delay i.e., $\frac{\text{distance}}{\text{speed}} = \text{time}$ or time = $\frac{\text{distance}}{\text{speed}} = \frac{r}{u}$.
- The current distribution at the origin is expressed as-

$$I = I_0 \cos\omega\left(t - \frac{r}{u}\right)$$

Since we know that-

$\beta = \frac{\omega}{c}$ and u is the velocity.

$$\therefore I = I_0 \cos\left(\omega t - \frac{\omega r}{u}\right)$$

$$I = I_0 \cos(\omega t - \beta r)$$

$$I = I_0 \text{ Re } [e^{j\omega t} - e^{-j\beta r}]$$

Rack your Brain

In radio wave systems or microwave links, the transmission loss per unit length decreases with _____ distance.

- Consider components $A_x = A_y = 0$ and suppressing the time factor, then we get-

$$\vec{A}_{zs} = \frac{\mu I_0 e^{-j\beta r} dl}{4\pi r} \hat{a}_z$$

- The required coordinate system is spherical, but the above magnetic vector potential is expressed in the cartesian system.
Now, converting into a spherical coordinate system, we know that-
Here, 's' the subscript is used to just denote that the time factor is suppressed.

$$\begin{bmatrix} A_{rs} \\ A_{\theta s} \\ A_{\phi s} \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \\ \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta \\ -\sin\phi\cos\phi & \cos\phi & 0 \end{bmatrix} \begin{bmatrix} A_{xs} \\ A_{ys} \\ A_{zs} \end{bmatrix}$$

- As we know that-

$x = r \sin\theta \cos\phi$

$y = r \sin\theta \sin\phi$

$z = r \cos\theta.$

$A_{xs} = A_{ys} = 0$ but $A_{zs} \neq 0$

- By solving we get-

  $A_{rs} = A_{zs} \cos\theta$

  $A_{\theta s} = -A_{zs} \sin\theta$

  $A_{\phi s} = 0$

  The first step has been followed as spherical coordinate has been analyzed.
- Now, calculating $\vec{H}$ and $\vec{B}$ as follows-

  We know, for free space

  $$\vec{B}_s = \mu_0 \vec{H}_s = \nabla \times \vec{A}_s$$

  $$\vec{B}_s = \mu_0 \left[H_{rs}\hat{a}_r + H_{\theta s}\hat{a}_\theta + H_{\phi s}\hat{a}_\phi \right]$$

  $$\vec{B}_s = \frac{1}{r^2 \sin\theta} \begin{bmatrix} \hat{a}_r & -r\hat{a}_\theta & r\sin\theta\hat{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_{rs} & rA_{\theta s} & r\sin\theta A_{\phi s} \end{bmatrix} = \frac{1}{r^2 \sin\theta} \begin{bmatrix} \hat{a}_r & -r\hat{a}_\theta & r\sin\theta\hat{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_{rs} & rA_{\theta s} & 0 \end{bmatrix}$$

  Since, $A_{\phi s} = 0$
- Comparing 'r' components on both sides, we get-

  $$\mu_0 H_{rs} = \frac{1}{r^2 \sin\theta} \left[\frac{\partial}{\partial \theta}(0) - \frac{\partial}{\partial \phi}(rA_{\theta s}) \right] \quad \left\{ \because \ A_{\phi s} = 0 \right\}$$

  $$\mu_0 H_{rs} = \frac{-1}{r^2 \sin\theta} \frac{\partial}{\partial \phi} \left[-rA_{zs} \sin\theta \right]$$

  $$\mu_0 H_{rs} = \frac{1}{r^2 \sin\theta} \frac{\partial}{\partial \phi} \left[r\sin\theta \frac{\mu_0 I_0 dl e^{-j\beta r}}{4\pi r} \right]$$

  $\mu_0 H_{rs} = 0$ or $H_{rs} = 0$

- It is concluded that $H_{rs} = 0$

  Similarly, we will get $H_{\theta s}$ and $H_{\phi s}$ as well.
- On comparing θ-components, we get-

  $$\mu_0 H_{\theta s} = \frac{-r}{r^2 \sin\theta} \left[\frac{\partial}{\partial r}(0) - \frac{\partial}{\partial \phi}(A_{rs}) \right]$$

Gray Matter Alert!!!

In coaxial cables or waveguides, transmission loss per unit length generally increases with respect to the frequency.

$$\mu_0 H_{\theta s} = \frac{1}{r \sin\theta} \frac{\partial}{\partial \phi} [A_{zs} \cos\theta]$$

$$\mu_0 H_{\theta s} = \frac{1}{r \sin\theta} \frac{\partial}{\partial \phi} \left[\frac{\mu_0 I_0 dl e^{-j\beta r}}{4\pi r} \cos\theta \right]$$

Thus, $H_{\theta s} = 0$

- Now, comparing ϕ-component on both sides, we get-

$$\mu_0 H_{\phi s} = \frac{r \sin\theta}{r^2 \sin\theta} \left[\frac{\partial}{\partial r} (r A_{\theta s}) - \frac{\partial}{\partial \theta} (A_{rs}) \right]$$

$$\mu_0 H_{\phi s} = \frac{1}{r} \left[\frac{\partial}{\partial r} (-r A_{zs} \sin\theta) - \frac{\partial}{\partial \theta} (A_{zs} \cos\theta) \right]$$

$$\mu_0 H_{\phi s} = \frac{1}{r} \left[\frac{-\partial}{\partial r} \left(r \frac{\mu_0 I_0 dl e^{-j\beta r}}{4\pi r} \sin\theta \right) - \frac{\partial}{\partial \theta} \left(\frac{\mu_0 I_0 dl e^{-j\beta r}}{4\pi r} \cos\theta \right) \right]$$

$$\mu_0 H_{\phi s} = \frac{1}{r} \left[- \frac{\mu_0 I_0 dl (-j\beta) e^{-j\beta r} \sin\theta}{4\pi} + \frac{\mu_0 I_0 dl e^{-j\beta r} \sin\theta}{4\pi r} \right]$$

$$\mu_0 H_{\phi s} = \frac{\mu_0 I_0 dl e^{-j\beta r}}{4\pi} \left[\frac{j\beta}{r} + \frac{1}{r^2} \right]$$

Hence, $H_{\phi s} = \frac{\mu_0 I_0 dl e^{-j\beta r}}{4\pi} \left[\frac{j\beta}{r} + \frac{1}{r^2} \right]$

and $H_{rs} = 0$ and $H_{\theta s} = 0$

- The magnetic field at far-field is defined by the above equation while neglecting $\frac{1}{r^2}$ terms in the above equation.
- For calculating electric field components.
  Using Ampere's law, we know that in the spherical coordinate system-

$$\nabla \times \vec{H}_s = (\sigma + j\omega \in) \vec{E}_s$$

Since, $\sigma = 0$(lossless)

$$\nabla \times \vec{H}_s = j\omega \in \left[E_{rs} \hat{a}_r + E_{\theta s} \hat{a}_\theta + E_{\phi s} \hat{a}_\phi \right]$$

$$j\omega \in \left[E_{rs}\hat{a}_r + E_{\theta s}\hat{a}_\theta + E_{\phi s}\hat{a}_\phi\right] = \frac{1}{r^2 \sin\theta}\begin{vmatrix} \hat{a}_r & -r\hat{a}_\theta & r\sin\theta\hat{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial\theta} & \frac{\partial}{\partial\phi} \\ H_{rs} & H_{\theta s} & r\sin\theta H_{\theta s} \end{vmatrix}$$

Since we know that, $E_{\phi s} = 0$
$H_{rs} = 0$ and $H_{\theta s} = 0$ and substituting the value of H_{fs}, we get-

$$j\omega\epsilon_0 E_{rs} = \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial\theta}\left[r\sin\theta \frac{I_0 dl \cdot \sin\theta}{4\pi} e^{-j\beta r}\left(\frac{j\beta}{r} + \frac{1}{r^2}\right)\right]$$

$$j\omega\epsilon_0 E_{rs} = \frac{1}{r^2 \sin\theta}\frac{rI_0 dl 2\sin\theta\cos\theta e^{-j\beta r}}{4\pi}\left(\frac{j\beta}{r} + \frac{1}{r^2}\right)$$

$$E_{rs} = \frac{I_0 dl\cos\theta e^{-j\beta r}}{2\pi r(j\omega\epsilon_0)}\left(\frac{j\beta}{r} + \frac{1}{r^2}\right)$$

$$E_{rs} = \frac{I_0 dl\cos\theta e^{-j\beta r}}{2\pi(j\omega\epsilon_0)}(j\beta)\left(\frac{1}{r^2} + \frac{1}{j\beta r^3}\right)$$

As we know that-

$$\beta = \omega\sqrt{\mu_0\ \epsilon_0}$$

and $\eta = \sqrt{\frac{\mu_0}{\epsilon_0}}$

$$\therefore \frac{\beta}{\eta} = \frac{\omega\sqrt{\mu_0\ \epsilon_0}}{\sqrt{\frac{\mu_0}{\epsilon_0}}} = \omega\epsilon_0$$

or $\eta = \frac{\beta}{\omega\epsilon_0}$

- Now substituting the η value in E_{rs} equation, we get-

$$E_{rs} = \frac{I_0 dl\cos\theta e^{-j\beta r}\eta}{2\pi}\left(\frac{1}{r^2} - \frac{1}{\beta r^3}\right)$$

Rack your Brain

An antenna acts as a sensor of electromagnetic waves. (Yes/No)

Previous Year's Questions

An antenna, when radiating, has a highly directional radiation pattern. When the antenna is receiving, Its radiation pattern-
(A) is more directive
(B) is less directive
(C) is the same
(D) exhibits no directivity

- For deriving $E_{\theta s}$, we compare θ-components and we get-

$$j\omega \in_0 E_{\theta s} = \frac{-1}{r^2 \sin\theta} r \frac{\partial}{\partial r}\left[r \sin\theta \frac{I_0 dl \sin\theta e^{-j\beta r}}{4\pi}\left(\frac{j\beta}{r} + \frac{1}{r^2}\right)\right]$$

$$E_{\theta s} = \frac{-1}{(j\omega \in_0) r} \frac{I_0 dl \sin\theta}{4\pi} \frac{\partial}{\partial r}\left[e^{-j\beta r}(j\beta) + \frac{e^{-j\beta r}}{r}\right]$$

$$E_{\theta s} = \frac{-I_0 dl \sin\theta}{(j\omega \in_0) 4\pi r}\left[-(j\beta \times j\beta) e^{-j\beta r} + e^{-j\beta r}\left(\frac{-1}{r^2}\right) + \frac{1}{r}(-j\beta) e^{-j\beta r}\right]$$

$$E_{\theta s} = \frac{-I_0 dl \sin\theta e^{-j\beta r}}{(j\omega \in_0)(4\pi r)}\left[\beta^2 - \frac{1}{r^2} - \frac{j\beta}{r}\right]$$

$$E_{\theta s} = \frac{-I_0 dl \sin\theta e^{-j\beta r}}{4\pi j}\left(\frac{\beta}{\omega \in_0}\right)\left[\frac{\beta}{r} - \frac{1}{\beta r^3} - \frac{j}{r^2}\right]$$

$$E_{\theta s} = \eta \frac{I_0 dl \sin\theta e^{-j\beta r}}{4\pi}\left[\frac{j\beta}{r} - \frac{j}{\beta r^3} + \frac{1}{r^2}\right]$$

$$E_{\theta s} = \eta \frac{I_0 dl \sin\theta e^{-j\beta r}}{4\pi}\left[\frac{j\beta}{r} + \frac{j}{r^2} - \frac{1}{j\beta^3}\right]$$

Rack your Brain

Receiving antennas in radio receivers normally are ferrite or loop antenna. (Yes/No)

- For Hertzian dipole radiation fields are-

$E_{\phi s} = 0$, $H_{rs} = 0$, $H_{\phi s} = 0$

$$H_{\phi s} = \frac{I_0 dl \sin\theta e^{-j\beta r}}{4\pi}\left[\frac{j\beta}{r} + \frac{1}{r^2}\right]$$

$$E_{rs} = \eta \frac{I_0 dl \sin\theta e^{-j\beta r}}{2\pi}\left[\frac{1}{r^2} - \frac{j}{\beta r^3}\right]$$

$$E_{\theta s} = \eta \frac{I_0 dl \sin\theta e^{-j\beta r}}{4\pi}\left[\frac{j\beta}{r} + \frac{1}{r^2} - \frac{j}{\beta r^3}\right]$$

Previous Year's Questions

A transverse electromagnetic wave with circular polarization is received by a dipole antenna. Due to polarization mismatch, the power transfer efficiency from the wave to the antenna is reduced to about-

(A) 50% (B) 35.3%
(C) 25% (D) 0%

- For far-field-

(i) $\left(\frac{1}{r}\right)$ term is called far-field or radiation field.

i.e., if r >> very large.

and $\left(\frac{1}{r^2}\right)$ & $\left(\frac{1}{r^3}\right)$ will be very small or tends to have zero value.

Thus, above equation will be modified by applying above given condition of far-field.

Therefore, we get-

$E_{\phi s} = E_{rs} = H_{rs} = H_{\theta s} = 0$

and $H_{\phi s} = \frac{I_0 dl \sin\theta e^{-j\beta r}}{4\pi r}(j\beta)$

and $E_{\theta s} = \eta \frac{I_0 dl \sin\theta e^{-j\beta r}}{4\pi r}(j\beta)$

- Also we know that-
  Intrinsic impedance η is expressed as-

$$\eta = \frac{E_{\theta s}}{H_{\theta s}}$$

- As we need to draw radiation pattern i.e., electric field pattern.
  Thus, $E_{\theta s}$ equation is considered to be an important equation.

(ii) The $\left(\frac{1}{r^2}\right)$ term is referred to as induction field.

(iii) And $\left(\frac{1}{r^3}\right)$ term is called near-field or electrostatic field.

(iv) The distance at which magnitude of far-field and near-field are equal to called as radian-distance i.e., the magnitude of $\left(\frac{1}{r}\right)$ term and magnitude of $\left(\frac{1}{r^3}\right)$ term in equal.
For this,

$$\left|\frac{j\beta}{r}\right| = \left|-\frac{j}{\beta r^3}\right|$$

$$\frac{\beta}{r} = \frac{1}{\beta r^3}$$

Rack your Brain

Does the antenna act as an impedance matching device?

Rack your Brain

The units of radiation intensity are _____.

Previous Year's Questions

A 1 km long microwave link uses two antennas each having 30 dB gain. If the power transmitted by one antenna is 1 W at 3 GHz, the power received by the other antenna is approximately-

(A) 98.6 μW (B) 76.8 μW
(C) 63.4 μW (D) 55.2 μW

$$r = \frac{1}{\beta} = \frac{\lambda}{2\pi}$$

At $r \simeq \frac{\lambda}{6}$ $\{\because 2\pi \simeq 6\}$

Thus, at $r \simeq \frac{\lambda}{6}$, far-field and near fields are equal and it is defined as radian distance.

(v) The boundary between far-field and near field is given by-

$$r \geq \frac{2d^2}{\lambda}$$

or $\begin{cases} r >> \frac{2d^2}{\lambda} \rightarrow \text{Far} - \text{field} \\ r < \frac{2d^2}{\lambda} \rightarrow \text{Near} - \text{field} \end{cases}$

where, d = largest dimension of the antenna

λ = wavelength

- At far-field,

$$E_{\theta s} = \eta \frac{I_0 dl \sin\theta e^{-j\beta r}}{4\pi r} j\beta$$

$$H_{\phi s} = \frac{E_{\theta s}}{\eta}$$

The power radiated will be-

$$P_{rad} = \int_s \vec{P}_{avg} \cdot d\vec{S}$$

As we know that-

$$\vec{P}_{avg} = \frac{1}{2} Re\left[\vec{E}_s \times \vec{H}_s^*\right] = \frac{1}{2} Re\left[E_{\theta s}\hat{a}_\theta \times H_{\phi s}\hat{a}_\phi\right]$$

$$\vec{P}_{avg} = \frac{1}{2} Re\left[\eta H_{\phi s}\hat{a}_\theta \times H_{\phi s}^*\hat{a}_\phi\right]$$

$$\vec{P}_{avg} = \frac{1}{2} Re\left[\eta H_{\phi s} H_{\phi s}^* \left(\hat{a}_\theta \times \hat{a}_\phi\right)\right]$$

Previous Year's Questions

For a dipole antenna -

(A) The radiation intensity is maximum along the normal to the dipole axis.

(B) The current distribution along its length is uniform irrespective of the length.

(C) The effective length equals its physical length.

(D) The input impedance is independent of the location of the feed-point.

Rack your Brain

The effective length of a wire is always greater than the actual length. (Yes/No)

$$\vec{P}_{avg} = \frac{1}{2}\left[\eta \,|\, H_{\phi s} \,|^2 \; \hat{a}_r\right] \qquad \left\{\because H_{\phi s} H_{\phi s}^{*} = |H_{\phi s}|^2\right\}$$

$$\therefore P_{rad} = \int_s \vec{P}_{avg} \cdot d\vec{S}$$

$$P_{rad} = \int_s \frac{1}{2}\eta \,|\, H_{\phi s} \,|^2 \; \hat{a}_r \cdot r^2 \sin\theta \, d\theta \, d\phi \; \hat{a}_r$$

$$P_{rad} = \frac{1}{2}\eta \int_s |\, H_{\phi s} \,|^2 \; r^2 \sin\theta \, d\theta \, d\phi$$

$$P_{rad} = \frac{1}{2}\eta \int_s \left|\frac{E_{\theta s}}{\eta}\right|^2 r^2 \sin\theta \, d\theta \, d\phi$$

$$P_{rad} = \int_s \frac{\eta^2 I_0^2 dl^2 \sin^2\theta \beta^2}{(4\pi)^2 (r^2)(2\eta)} r^2 \sin\theta \, d\theta \, d\phi$$

$$P_{rad} = \frac{(120\pi)^2 I_0^2 dl^2 \beta^2}{16\pi^2 \times 2 \times 120\pi} \int_{\phi=0}^{2\pi} d\phi \int_{\theta=0}^{\pi} \sin^3\theta \, d\theta$$

$$P_{rad} = \frac{120\pi I_0^2 dl^2 (2\pi)^2}{16\pi \times 2(\lambda)^2} \times 2\pi \times 2\int_{\theta=0}^{\pi/2} \sin^3\theta \, d\theta \qquad \left[\because \beta = \frac{2\pi}{\lambda}\right]$$

$$P_{rad} = \frac{60\pi^2 I_0^2 dl^2 (2\pi)^2}{\lambda^2}\left(\frac{2}{3}\right) = 40\pi^2 I_0^2 \left(\frac{dl}{\lambda}\right)^2$$

$$P_{rad} = 40\pi^2 I_0^2 \left(\frac{dl}{\lambda}\right)^2$$

where I_0 is the peak current.

- Radiation resistance will be expressed as-

$$P_{rad} = I_{rms}^2 R_{rad}$$

$$P_{rad} = \frac{I_0^2}{2} R_{rad}$$

$$P_{rad} = \frac{2P_{rad}}{I_0^2} = \frac{2}{I_0^2} 40\pi^2 I_0^2 \left(\frac{dl}{\lambda}\right)^2$$

Previous Year's Questions

A parabolic dish antenna has a conical beam 2° wide. The directivity of the antenna is approximately-

(A) 20 dB (B) 30 dB
(C) 40 dB (D) 50 dB

Rack your Brain

The magnetic field is developed _____ an antenna.

$$\therefore P_{rad} = 80\pi^2 \left(\frac{dl}{\lambda}\right)^2$$

- For Hertzian dipole antenna we know that the length is-

$$\left(dl \leq \frac{\lambda}{10}\right)$$

If we assume, $dl = \frac{\lambda}{20}$

Then, $R_{rad} = 80\pi^2 \left(\frac{\lambda}{20\lambda}\right)^2 = \frac{80\pi^2}{400} \simeq 2\,\Omega$ (Very small value)

Now, radiated power will be-

$P_{rad} = I_{rms}^2 R_{rad}$

On decreasing radiated resistance, the radiated power will also decrease.
And we know that-

Antenna efficiency, $\eta = \frac{P_{rad}}{P_{input}}$

As P_{rad} decreases, the antenna efficiency will also decrease.
Thus, because of less efficiency, Hertzian dipole is not used widely.

Rack your Brain

The efficiency of an antenna in terms of directive and power gains is_____.

Previous Year's Questions

The radiation resistance of a circular loop of one turn is 0.01 Ω. The radiation resistance of five turns of such a loop will be _____.

(A) 0.002 Ω (B) 0.01 Ω
(C) 0.05 Ω (D) 0.25 Ω

Half-wave Dipole Antenna

- The half-wave dipole antenna, as its name implies, has a length equal to half of its wavelength.
- This antenna is made up of thin wire that is excited in the middle by a voltage source connected via a transmission line to the antenna.

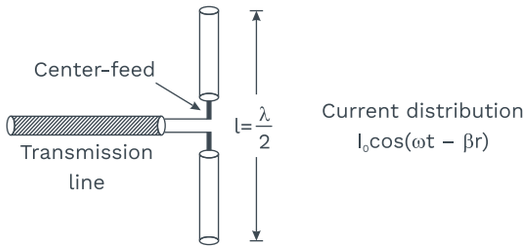


Figure 8.4: Half-wave dipole antenna

- For half-wave dipole, the magnetic vector potential is given by-

$$A_{zs} = \frac{\mu I_0 \cos\left(\frac{\pi}{2}\cos\theta\right) e^{-j\beta r}}{2\pi r\beta \sin^2\theta}$$

- At far-field, the radiation field patterns-

$$H_{\theta s} = \frac{jI_0 \cos\left(\frac{\pi}{2}\cos\theta\right) e^{-j\beta r}}{2\pi r \sin\theta}$$

and $E_{\theta s} = \eta H_{\theta s}$

- The radiated power is expressed as-

$$P_{rad} = 36.56 \times I_0^2$$

And the radiation resistance is given by-

$$R_{rad} = 2\frac{P_{rad}}{I_0^2} = 73\,\Omega \text{ (Constant value)}$$

$$\therefore (P_{rad})_{\text{Half-wave}} > (P_{rad})_{\text{Hertzian}}$$

Thus, the more power is radiated by the half-wave dipole antenna.

QUARTER-WAVE MONOPOLE ANTENNA

- For this type of antenna, the length of the element is required to be $l = \frac{\lambda}{4}$.

The source feeding is at the base of the antenna element.

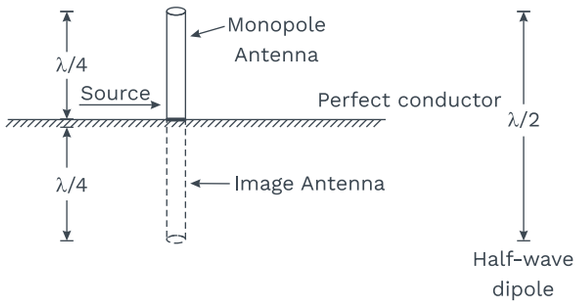


Figure 8.5: Quarter wave monopole antenna

Rack your Brain

The radiation fields are nothing but far-fields. (True/ False)

Previous Year's Questions

An antenna in free space receives 2 μW of power when the incident electric field is 20 mV/m rms. The effective aperture of the antenna is-

(A) 0.005 m^2 (B) 0.05 m^2
(C) 1.885 m^2 (D) 3.77 m^2

- Due to the property of perfect conductor property i.e., the same length of an antenna placed above perfect conductor will be formed below it and is referred to as an image antenna.
  Hence, the total length of the antenna becomes $\frac{\lambda}{2}$ and it behaves as the half-wave dipole antenna.
  It can be noted that feeding is at a center like a half-wave dipole antenna.
  Therefore, the quarter-wave monopole antenna-

  $R_{rad} = 36.5\ \Omega$

  and $P_{rad} = 36.5 I_0^2$

SMALL LOOP ANTENNA

- A loop antenna is a RF current carrying radiating coil with an appropriate cross-section of one or more radiating loops.
- Loop antennas have an advantage over other conventional antennas (as they are small in size) and can be used in a variety of mobile communication applications. The loop antenna's polarization is identical to that of short dipole.
- For the far-field, the field pattern equations are as follows-

  $$E_{\phi s} = \frac{120\pi^2 I_0}{r}\frac{S}{\lambda^2}\sin\theta\, e^{-j\beta r} \text{ and } H_{\theta s} = -\frac{E_{\phi s}}{\eta}$$

- The radiation resistance is expressed as-

  $$R_{rad} = \frac{320\pi^4 S^2}{\lambda^4}$$

  or $R_{rad} \propto S^2$

Rack your Brain

The far-field is implied by the presence of_____.

Previous Year's Questions

A transmitting antenna radiates 251 W isotopically. A receiving antenna, located 100 m away from the transmitting antenna, has an effective aperture of 500 cm^2. The total power received by the antenna is-
(A) 10 μW (B) 1 μW
(C) 20 μW (D) 100 μW

where S = Cross-sectional area

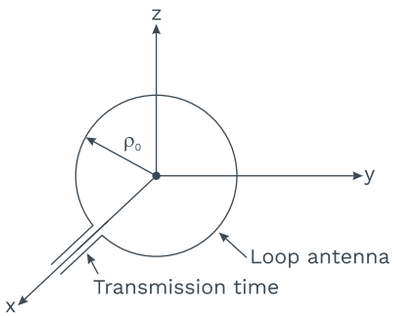


Figure 8.6: Small loop antenna

here ρ_0 << radius of the loop (∴ Small loop)

- For square loop of side 'a' -

  $S = a^2$

  And if loop is having N turns then,

  $S = Na^2$

- For the large loop ($C \geq 3.14\lambda$) antenna

  $$R_{rad} = 60\pi^2 \left(\frac{C}{\lambda}\right)$$

  Radiation resistance of loop antenna:

  $$R_{rad} = 20\pi^2 \left(\frac{C}{\lambda}\right)^2 \left(\frac{\mu_\theta}{\mu_0}\right) N^2$$

  Here, C is the circumference of the large loop ($C = 2\pi r$).

Example 1:
Consider a Hertzian dipole antenna that has a length of 4.6 mm and carries a 60 MHz frequency and current of 2.8 A. Let the magnitude of the electric field is determined at distance 'x' from the antenna. Thus, it will be considered as far-field if the distance x is equal to-

Rack your Brain

What are the applications of the Yagi-Uda antenna?

Rack your Brain

The induction-field is demonstrated by the presence of_____.

(A) 8.3 mm (B) 8.04 mm
(C) 8.4 mm (D) 8 mm

Solution:
The far-field is considered if the distance is given as-

$$r \geq \frac{2d^2}{\lambda}$$

Here, given that d = 4.6 mm = 4.6×10^{-3} m

Frequency, f = 60 MHz

Current, I = 2.8 A
By substituting the values, we get-

$$x \geq \frac{2(4.6 \times 10^{-3})^2}{\left(\frac{3 \times 10^8}{60 \times 10^6}\right)}$$

$$x \geq \frac{42.32 \times 10^{-6}}{5}$$

$$x \geq 8.46 \times 10^{-6}$$

Thus, the distance x must be greater or equal to 8.464 mm.
Hence, the correct option is (C).

Example 2:
If a dipole 5 m long carries a 4A current and operates at 5MHz. The average radiated power (in W) is-

Solution:
Given-

Length, l = 5 m

Current, I = 4 A

Frequency, f = 5 MHz

The wavelength (λ) is given by-

Previous Year's Questions

If the diameter of a $\frac{\lambda}{2}$ dipole antenna is increased from $\frac{\lambda}{100}$ to $\frac{\lambda}{50}$, then its-

(A) bandwidth increases
(B) bandwidth decreases
(C) gain increases
(D) gain decrease

Rack your Brain

The radiation resistance (R_{rad}) of an isolated half-wave dipole is_____.

$$\lambda = \frac{c}{f} = \frac{3\times10^8}{5\times10^6} = 60\,m$$

Since, $l \le \frac{\lambda}{10}$ (for Hertzian dipole)

$$l \le \frac{60}{10}$$

$$\therefore l \le 6\ m$$

Given, l = 5 m

∴ Given that the antenna is a Hertzian dipole antenna.

The power radiated will be-

$$P_{rad} = \frac{I_0^2}{2} R_{rad}$$

and $R_{rad} = 80\pi^2\left(\frac{dl}{\lambda}\right)^2 = 80\pi^2\left(\frac{5}{60}\right)^2 = 5.48\,\Omega$

$$P_{rad} = \frac{(4)^2}{2}\times 5.48$$

$$P_{rad} = 43.86\ W$$

Hence, the correct option is (A).

Example 3:

Consider a car's radio antenna of length 2m operating in the AM frequency of 2 MHz. The current required to transmit 3.4 W of power is-

(A) 4.31 (B) 6.96

(C) 21.13 (D) 48.57

Solution:

The frequency is 2 MHz

Then, the wavelength will be-

$$\lambda = \frac{c}{f} = \frac{3\times10^8}{2\times10^6} = 150\,m$$

As the given length l (i.e., 2m) is less than the $\frac{\lambda}{10}$. Thus, the given antenna is a Hertzian antenna.

Rack your Brain

The Hertzian antenna operates on _____ of its fundamental frequency.

Rack your Brain

The radiation resistance of a quarter-wave monopole is _____ ohms.

$\therefore l \leq \frac{\lambda}{10}$

Here, $2\,m \leq \frac{150}{10}$

$2 \leq 15$ (Condition satisfied)

Thus, the given antenna is Hertzian dipole antenna

$\therefore$ Radiation resistance (R_{rad}) is given by-

$$R_{rad} = 80\pi^2\left(\frac{dl}{\lambda}\right)^2$$

$$= 80\pi^2\left(\frac{2}{150}\right)^2 = 0.14\Omega$$

Radiating power, $P_{rad} = \frac{1}{2}I_0^2 R_{rad}$

Current, $I_0 = \sqrt{\frac{2P_{rad}}{R_{rad}}} = 6.96A$

Hence, the correct option is (B).

Example 4:

Consider a circular loop antenna with a mean radius of 1.5 cm and N turns. If this given antenna is operating at 100 MHz. How many turns are required to produce a radiation resistance of 10 Ω?

(A) 230 (B) 228

(C) 520 (D) 200

Solution:

Given that-

The radius of the circular loop = 1.5 cm

$r = 1.5 \times 10^{-2}$ m

Frequency, f = 100 MHz = 100×10^6 Hz

and R_{rad} = 10 W

The radiation resistance (R_{rad}) is given as-

Rack your Brain

The current distribution in a half-wave dipole is_____.

Rack your Brain

The directive gain decreases as the antenna length increases. (True/False)

$$R_{rad} = \frac{320\pi^4 S^2}{\lambda^4}$$

For circular loop for N turns,

$$R_{rad} = \frac{320\pi^4 (N\pi r^2)^2}{\lambda^4}$$

$$\frac{R_{rad} \times \lambda^4}{320\pi^4} = N^2\pi^2 r^4$$

$$N = \sqrt{\frac{R_{rad} \times \lambda^4}{320\pi^6 r^4}}$$

and $\lambda = \frac{c}{f} = \frac{3 \times 10^8}{100 \times 10^6} = 3\,m$

By substituting all values, we get -

$$\therefore N = \sqrt{\frac{10 \times (3)^4}{320\pi^6 (1.5 \times 10^{-2})^4}}$$

$$N = \sqrt{\frac{10 \times 81}{320\pi^6 (1.5 \times 10^{-2})^4}}$$

$$N = 228.05 \approx 228$$

Hence, the correct option is (B).

Example 5:

If the loop antenna is built of copper wire with a radius of 5 mm and has loop radius of 0.5 m. The loop radiates at 6 MHz frequency, a current of 40 A, and conductivity of 5.8 × 10^7S. Evaluate the radiation resistance (in mΩ), power radiated (in W) and radiation efficiency (in %) respectively,-

(A) 6.14, 2.13, 2.1 (B) 2.46, 4.13, 20.15
(C) 5.39, 4.312, 37.1 (D) 6.13, 4.312, 30.19

Solution:

Given,

Loop radius = 0.5 m

Copper wire's radius = 5 mm = 5 × 10^{-3} m

Rack your Brain

The current distribution in an alternating current element is constant. (True/ False)

Rack your Brain

Is the number of nulls in the resultant pattern is equal to the sum of the nulls in the individual pattern and array pattern?

Frequency (f) = 6.5 MHz = 6.5×10^6 Hz
Current (I) = 40 A
The radiation resistance is expressed as-

$$R_{rad} = \frac{320\pi^4 S^2}{\lambda^4}$$

$$= \frac{320 \times \pi^4 \times \pi(0.5)^2}{\left(\frac{3 \times 10^8}{6.5 \times 10^6}\right)}$$

$$= \frac{320\pi^5(0.25)}{(46.15)^4}$$

$\therefore R_{rad} = 5.39$ mW

The power radiated is given by-

$$P_{rad} = \frac{I_0^2}{2} R_{rad}$$

$$P_{rad} = \frac{(40)^2}{2} \times 5.39 \times 10^{-3}$$

$\therefore P_{rad} = 4.312$ W

The radiation efficiency is given by-

$$\eta = \frac{R_{rad}}{R_{rad} + R_l}$$

where $R_l = \frac{l}{2\pi a}\sqrt{\frac{\pi f \mu_0}{\sigma}}$

$$R_l = \frac{0.5}{2\pi \times 5 \times 10^{-3}}\sqrt{\frac{\pi \times 6.5 \times 10^6 \times 4\pi \times 10^{-7}}{5.8 \times 10^7}} = \frac{1}{2\pi \times 10^{-2}} \times 6.39 \times 10^{-4}$$

$R_l = 1.017 \times 10^{-2}$

Radiation efficiency, $\eta = \frac{5.39 \times 10^{-3}}{5.39 \times 10^{-3} + 1.017 \times 10^{-2}} = 0.371 = 37.1\%$

Hence, the correct option is (C).

Rack your Brain

The radiation patterns of horizontal and vertical dipoles are identical. (True/ False)

Example 6:
If the radiation resistance of a circular loop is 0.02Ω. Evaluate the radiation resistance of seven turns of the same loop will be-
(A) 2 Ω (B) 0.98 Ω
(C) 0.02 Ω (D) 1 Ω

Solution:
Given,
The radiation resistance of a circular loop $(R_{rad}) = 0.02\ \Omega$
As we know that, $R_{rad} \propto N^2$

$$\frac{(R_{rad})_1}{(R_{rad})_2} = \left(\frac{N_1}{N_2}\right)^2$$

$$\frac{0.02}{(R_{rad})_2} = \left(\frac{1}{7}\right)^2 \text{ (Given } N_2=7)$$

$$(R_{rad})_2 = 0.98\ \Omega$$

Hence, the correct option is (B).

Rack your Brain

Is the dummy antenna a radiating antenna?

Example 7:
A wave radio transmitter operating at a wavelength of 479 m has a tower antenna of height 120 m. Determine the radiation resistance of the antenna.
(A) 36.5 Ω (B) 73 Ω
(C) 37 Ω (D) 30.5 Ω

Solution:
Given-
Wavelength $(\lambda) = 479$ m
Length of antenna, $l = 120$ m
and here, $l \simeq \frac{\lambda}{4} = \frac{479}{4} \simeq 120\,m$
Thus, given antenna is a quarter-wave antenna.
The radiation resistance of this antenna will be-

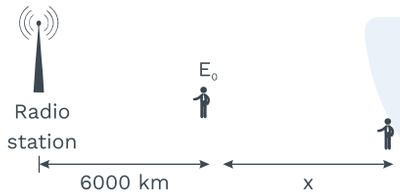
Rack your Brain

The current distribution in very short dipoles is triangular. (True/ False)

R_{rad} = 36.5 Ω (constant value)
Hence, the correct option is (A).

Example 8:
If a person with the receiver is 6 km away from the transmitter. Upto what distance this person should move so as to detect a 3dB decrease in signal strength?
(A) 2131 m (B) 2000 m
(C) 2936 m (D) 2485.28 m

Solution:



As we know that-

$$\frac{E_0}{\sqrt{2}} = -3\,dB$$

$$20\log\frac{1}{\sqrt{2}} = -3\,dB$$

and for far-field, $E \propto \frac{1}{r}$

$$\frac{E_1}{E_2} = \frac{r_2}{r_1}$$

$$\frac{E_0}{(E_0/\sqrt{2})} = \frac{r_2}{r_1} = \frac{6000 + x}{6000}$$

Distance, x = 2485.28 m

Example 9:
For an antenna radiating in free space, the electric field at a distance of 2 km is found to be 14 mV/m. The magnitude of average power density due to this antenna at a distance of 4 km from the antenna is _____.

Rack your Brain

Patterns of Half-wave dipole and quarter-wave monopole are identical. (True/ False)

Solution:

Given,

Distance d_1 = 2 km

Electric field, E_1 = 14 mV/m

Distance d_2 = 3 km

The power density is given by-

$$\text{Power density} = \frac{\text{Power}}{\text{Area}} = \frac{1}{2\eta}E_0^2$$

Since, $E \propto \frac{1}{r}$

$$\frac{E_1}{E_2} = \frac{r_2}{r_1}$$

E_2 = 7 mV/m

$\therefore$ Power density at distance 4 km will be-

$$= \frac{1}{2 \times 120\pi} \times (7 \times 10^{-3})^2 = 6.49 \times 10^{-8} = 0.649$$ nW/m^2

ANTENNA CHARACTERISTICS

Radiation Pattern

- When the amplitude of a specified component of an electric field is plotted then, it is called field pattern or voltage pattern.
- As we know that radiation from the antenna can occur in all directions. Hence, it follows a spherical symmetry.
- An antenna following a spherical symmetry for radiation pattern will have 3 components i.e., (r, θ, ϕ) components. Thus, an electric field will have r, θ and ϕ to make a radiation pattern for the antenna.
- When the square of the amplitude of the electric field is plotted, then it is called a power pattern.

Rack your Brain

Radiation resistance of short monopole is_____Ω.

Previous Year's Questions

The half-power beam widths (HPBW) of an antenna in the two orthogonal planes are 100° and 60° respectively. The directivity of the antenna is approximately equal to-

(A) 2 dB (B) 5 dB

(C) 8 dB (D) 12 dB

- Normalized electric field versus θ for constant ϕ is referred to as E-plane pattern (vertical pattern).
- Normalized electric field versus ϕ for $\theta = \frac{\pi}{2}$ is referred as H-plane pattern (or azimuthal or horizontal pattern).
- Antenna pattern /Radiation pattern is a 3-D plot of its radiation at the far-field.
  Here, far-field means when an electric field or magnetic field is having only $\left(\frac{1}{r}\right)$ components or is inversely proportional to r.

  eg : $E_\theta = \frac{E_0}{4\pi r}$ (Far-field)

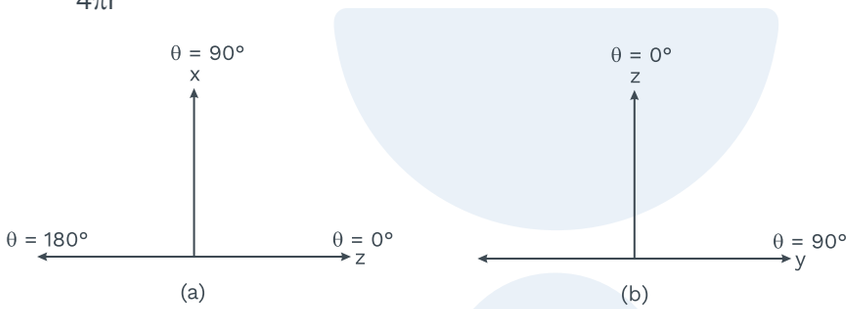


Figure 8.7: (a) E-plane (x-z plane) (b) E-plane (y-z plane)

- For the E-plane pattern, the z-axis should be there as $\theta = 0°$ at the z-axis.
  or
  As θ is determined from the z-axis and on varying θ and plot the electric field, that pattern is called the E-plane pattern.
- For $\theta = 90°$ H-plane is in z-direction and the plane will be x-y plane.

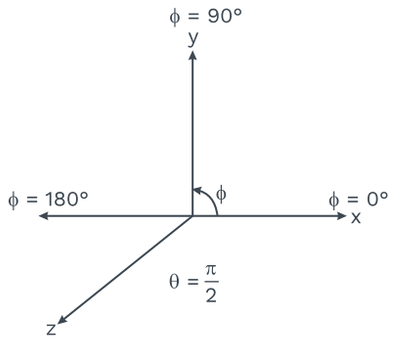


Figure 8.8: H-plane (x-y plane)

- In this case, varying ϕ in H-plane and ϕ is measured from the x-axis.

Note:

We know that, we take the z-axis for the E-plane whereas for the H-plane, we take x-axis as the principal axis. The y-axis is taken as the dummy axis.

- For normalizing the electric field –

$$E_{norm} = \frac{|\vec{E}|}{|\vec{E}|_{max}}$$

where $|\vec{E}|$ is the magnitude of $\vec{E}$.

- E-plane and H-plane are generally orthogonal.

Gain

- Gain is closely associated with directivity and directivity itself is dependent entirely upon the shape of the radiation pattern.
- Gain of Antenna may be defined as "The ratio of maximum radiation intensity in a given direction to the maximum radiation intensity from a reference antenna, produced in the same direction with the same power input where the effect of losses are involved in both test and reference antenna".
- When the reference antenna is taken as an isotropic antenna (100% efficiency).
- Gain of the subject antenna is denoted by G_0 and is known as gain with respect to an isotropic antenna.

$$G_0 = \frac{U'_{max}}{U_0}$$

U'_{max} =Maximum radiation from the subject antenna

U_0 = Radiation intensity from lossless antenna

Previous Year's Questions

For an 8 feet (2.4 m) parabolic dish antenna operating at 4 GHz, the minimum distance required for far-field measurement is closest to-

(A) 7.5 m (B) 15 m

(C) 20 m (D) 150 m

Rack your Brain

The radiated fields in the z-direction of the half-wave dipole consist of E_θ, E_r, H_θ terms only. (True/ False)

- Gain equals to directivity only when the antenna's efficiency is 100%.
- The gain in a direction from which the radiation intensity (U) is maximum may be designated by specifying the angle ϕ at which it is measured.

$$G_0(\theta,\phi)\ \frac{U_{max}}{U} = G_0$$

U_{max} = Maximum radiation intensity
U = Radiation intensity in direction of θ and ϕ

- Signal power received by a receiver at a distant point in the direction of maximum radiation, then the gain is defined as-

$$\text{Gain}(G) = \frac{\text{Maximum power received from gain antenna}(P_1)}{\text{Maximum power received from reference antenna}(P_2)}$$

$$G = \frac{P_1}{P_2}$$

Directive Gain

- The concentration of radiated power in any particular direction is defined as directive gain.
- Mathematically-

$$G_D(\theta,\phi) = \frac{4\pi U(\theta,\phi)}{P_{rad}}$$

$$G_D(\theta,\phi) = \frac{4\pi r^2 P_{avg}}{P_{rad}}$$

$$\therefore\ P_{avg} = \frac{G_D(\theta,\phi)P_{rad}}{4\pi r^2}$$

$$G_D(\theta,\phi) = \frac{4\pi r^2\left(\frac{E_0^2}{2\eta}\right)}{P_{rad}}$$

$$G_D(\theta,\phi) = \frac{4\pi r^2 E_0^2}{P_{rad}(2\eta)}$$

$$E_0 = \sqrt{\frac{G_D(\theta,\phi)P_{rad}(2\eta)}{4\pi r^2}}$$

- In terms of decibels, we get-
G_D (dB) = $10\log_{10} G_D(\theta, \phi)$

Rack your Brain

"The directivity of an antenna is determined by the beamwidth". (True/ False)

- Directive gain depends upon the distribution of radiated power in space. It does not depend on the input power of an antenna, antenna losses or power consumed in terminating resistance.
- Directive gain of an antenna in particular direction is defined as "the ratio of power density in a particular direction at a given distance to the power density that would be radiated at the same distance by an isotropic antenna.

Power Gain

- Power gain is the gain in which the ohmic loss is associated with the material.

$$G_p(\theta,\phi) = \frac{4\pi U(\theta,\phi)}{P_{in}}$$

where, $P_{in} = P_{rad} + P_{loss}$

$$P_{in} = I_{rms}^2 R_{rad} + I_{rms}^2 R_{loss}$$

$$P_{in} = \frac{I_0^2}{2}(R_{rad} + R_{loss})$$

And we know that, Loss resistance (R_l) is given by-

$$R_{loss} = \frac{l}{2\pi a}\sqrt{\frac{\pi f \mu}{\sigma}} = \frac{l}{2\pi a} \times R_s$$

where a = Radius of the conductor
L = length of the conductor
R_s = Surface resistance.

- In terms of power input, the power gain is defined as-

G_p (dB) = $10\log_{10} G_p$

$$G_p(dB) = 10\log_{10}\frac{P_1}{P_2}$$

Previous Year's Questions

A person with the receiver is 5 km away from the transmitter. What is the distance that this must move further to detect a 3-dB decrease in signal strength?

(A) 942 m (B) 2070 m
(C) 4978 m (D) 5320 m

Rack your Brain

If a current element is placed in the x-direction, the magnetic vector potential will be in the ____ direction.

Directivity

- It is defined as the maximum value of directive gain and denoted by D.
  In particular direction, the directivity (D) is a constant.
- Ratio of the maximum radiation intensity to its average radiation intensity is known as directivity.
  Mathematically-

$$D = \frac{U(\theta, \phi)_{max}}{U_{avg}}$$

- Directivity is also expressed by the ratio of maximum radiation intensity of subject antenna to radiation intensity of an isotropic or reference antenna (radiating the same total power).

$$D = \frac{U_{max}}{U_0}$$

$$D = \frac{U(\theta, \phi)_{max} \text{ (Test Antenna)}}{U(\phi_0) \text{ (Isotropic Antenna)}}$$

$$D = \frac{4\pi U_{max}}{P_{rad}}$$

- Average radiation intensity (U_{avg}) is obtained by dividing total power radiated by 4π radians.

$$D = \frac{U(\theta, \phi)_{max}}{P_{rad}/4\pi}$$

$$D = \frac{4\pi U(\theta, \phi)_{max}}{P_{rad}}$$

U_{max} = Maximum radiation intensity
U = Radiation intensity in direction of θ and ϕ

- In decibels: $D(dB) = 10\log_{10}D$
  For isotropic antenna, D = 1
  For Hertzian dipole antenna, D = 1.5

Rack your Brain

dB_m means power gain in dB compared to _____.

Previous Year's Questions

A medium wave radio transmitter operating at a wavelength of 492 m has a tower antenna of height 124m, the radiation resistance of the antenna is

(A) 25 Ω (B) 36.5 Ω
(C) 50 Ω (D) 73 Ω

For half-wave dipole antenna, D = 1.64

Also, $D = \frac{41253}{\theta_E^\circ \times \theta_H^\circ}$

where, θ_E° = HPBW in E-plane

and θ_H° = HPBW in H-plane.

Radiation Efficiency

The ratio of power radiated to total input power supplied to an antenna is termed as radiation efficiency of an antenna and it is denoted by η or K.

$$\eta = \frac{\text{Power Radiated}}{\text{Total Input Power}}$$

$$\eta = \frac{P_{rad}}{P_{Total}} = \frac{P_{rad}}{P_{rad} + P_{loss}}$$

where, P_{rad} = Power radiated, P_{loss} = Ohmic losses.

If the current flowing in the antenna is I, then-

$$\eta = \frac{\frac{I^2}{2} R_{rad}}{\frac{I^2}{2}(R_{rad} + R_{loss})}$$

$$\eta = \frac{R_{rad}}{R_{rad} + R_{loss}}$$

where, R_{rad} = Radiation resistance

R_{loss} = Ohmic loss of antenna conductor

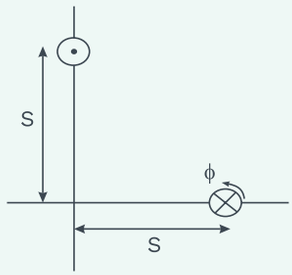
$\therefore$ R_{rad} + R_{loss} = Total effective resistance

- 'η' represents a fraction of total energy supplied to an antenna which is converted into electromagnetic waves.
- Antenna overall efficiency can be obtained as follows-

$$e_0 = e_{cd}\left(1 - |\Gamma|^2\right) = \frac{G_P}{G_D}$$

Previous Year's Questions

Two identical antennas are placed in the θ = π/2 plane as shown in the figure. The elements have equal amplitude excitation with 180° polarity difference, operating at a wavelength λ. The correct value of the magnitude of the far-zone resultant electric field, strength normalized with that of a single element, both computed for ϕ = 0° is-



(A) $2\cos\left(\frac{2\pi S}{\lambda}\right)$ (B) $2\sin\left(\frac{2\pi S}{\lambda}\right)$

(C) $2\cos\left(\frac{\pi S}{\lambda}\right)$ (D) $2\sin\left(\frac{\pi S}{\lambda}\right)$

Reflection efficiency, $e_r = \left(1 - |\Gamma|^2\right)$

And, $\Gamma = \dfrac{Z_{in} - Z_0}{Z_{in} + Z_0}$

where Z_{in}= Input impedance of the antenna.
Z_o= Characteristics impedance of transmission line.

- The gain of the antenna is given as-

  $G_P = e_{cd}\left(1 - |\Gamma|^2\right) G_D$

- If the antenna is lossless, R_{loss}= 0 then, e_{cd}= 1.
- If transmission line and the antenna are perfectly matched, i.e. $Z_{in} = Z_0$ then $\Gamma = 0$.

  And we can say that the power gain and directive gain will be the same i.e. $G_P = G_D$.

  If these conditions are not given, then by default we assume perfectly matched conditions.

Antenna Bandwidth

- Bandwidth (BW) over which gain is higher than some acceptable value.
- Bandwidth over which given front to back ratio (FBR) is achieved as-

$$\text{FBR} = \frac{\text{Power radiated in desired direction}}{\text{Power radiated in opposite direction}}$$

- Antenna bandwidth is a range of frequency above which the antenna maintains certain required characteristics like gain, FBR or SWR pattern (shape or direction), polarization and impedance.
  Basically, bandwidth of antenna depends on impedance and pattern.
  Let two frequency limits (i.e., ω_1 and ω_2)

  $\Delta\omega = \omega_2 - \omega_1$

  $BW = \dfrac{P_{rad}}{Q}$

  $\Delta\omega = \dfrac{P_{rad}}{Q}$

  $P_{rad} = 2\pi f_{rad}$

  $\Delta f = \dfrac{f_{rad}}{Q}$

$$\Delta f \propto \frac{1}{Q}$$

where f_{rad} = Centre / resonant / design frequency.

- The lower the 'Q' of the antenna, the higher the BW and vice versa.
- Frequency-independent antennas like log-periodic antennas (which has unlimited bandwidth) where lower and upper frequencies limits are specified independently. In those cases, bandwidth is represented by the ratio of the highest to the lowest operating frequency.
- Bandwidth (BW) is generally of low and moderate values are expressed in terms of the percentage of the center frequency.

$$BW\,\% = \frac{\text{Operating range}}{\text{Centre frequency}} \times 100$$

Antenna Beamwidth

- Antenna beamwidth is a measure of the directivity of an antenna.
- Antenna beamwidth is an angular width (in degrees) measured on radiation pattern (major lobe) between points where radiated power has fallen to half of its maximum value.
- This is termed as beamwidth between half-power points or half-power beamwidth (HPBW) because the power at half-power points is just half of its maximum value.
- HPBW is also known as 3dB beamwidth because, at half-power points, power is 3dB down the maximum power value of the major lobe.
- Half-power points the field intensity (i.e., voltage) equals $1/\sqrt{2}$ or 0.707 times its maximum value or 3dB down from maximum value.

Previous Year's Questions

A mast antenna consisting of a 50 meter long vertical conductor operates over a perfectly conducting ground plane. It is base fed at a frequency of 600 kHz. The radiation resistance of the antenna in ohms is-

(A) $\frac{2\pi^2}{5}$ (B) $\frac{\pi^2}{5}$

(C) $\frac{4\pi^2}{5}$ (D) $20\pi^2$

Rack your Brain

For a 200Ω antenna operating with 4A of current, the radiated power is _____. W

- Consider the radiation pattern in the below figure 8.9 and let P_1 and P_2 be the half-power points, the angle between P_1 and P_2 is the beamwidth of the antenna.
- The antenna beamwidth can be expressed as "the angular width (in degrees) of the major lobe between the two directions at which the radiated or received power is one-half the maximum power".

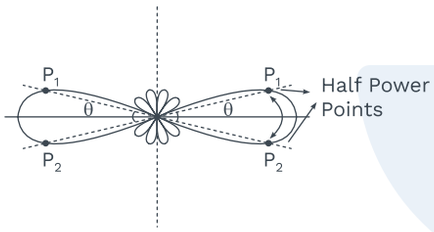


Figure 8.9: Beamwidth Angle P_1OP_2

- Radiation pattern of an angular width between first nulls or first side lobes, known as beamwidth between first nulls which is abbreviated as (BWFN) or beam width-10 dB down from maximum pattern.
- Directivity (D) is related to beam solid angle (Ω_A) (or beam area (BA)) as follows:

$$D = \frac{4\pi}{\Omega_A}$$

- Radiation pattern or lobe is a 3-dimensional, that has major lobe area given by the product of beamwidth in horizontal and vertical planes or E-plane and H-planes i.e.,
  BA ≃ (HPBW) in Horizontal plane × (HPBW) in the vertical plane (Radians)^2.
  BA ≃ (HPBW) in E Plane × (HPBW) in H plane.
  BA ≃ $\theta_E \times \theta_H$ (Radians)^2

Rack your Brain

The effective area of the antenna is a function frequency. (True/ False)

Previous Year's Questions

At 20 GHz, the gain of parabolic dish antenna of 1m diameter and 70% efficiency is-

(A) 15 dB (B) 25 dB
(C) 35 dB (D) 45 dB

$$D = \frac{4\pi}{\theta_E \times \theta_H} \text{ (Radians)}^2$$

$$D = \frac{4\pi \times (57.3)^2}{\theta_E^\circ \times \theta_H^\circ} \text{ (degree)}^2$$

{$\because$ 1 rad = 57.3°}

- Factor affecting beamwidth of an antenna are-
  (i) Shape of the radiation patterns
  (ii) Wavelengths
  (iii) Dimensions (e.g. radius of aperture)
- The beamwidth of the antenna is given in two planes perpendicular to each other when the beam is linearly polarized, E-plane and H-plane are used, called E-plane beamwidth and H-plane beamwidth.

Antenna Beam efficiency (BE)

- Antenna Beam efficiency is frequently used to judge the quality of transmitting & receiving antennas.
- Mathematically-

$$BE = \frac{\int_0^{2\pi}\int_0^{\theta} U(\theta,\phi)\sin\theta d\theta d\phi}{\int_0^{2\pi}\int_0^{\pi} U(\theta,\phi)\sin\theta d\theta d\phi}$$

here θ = Half-angle of the cone within which the percentage of total power is to be found.

- Beam efficiency (ε_M) is defined as the ratio of main beam area (Ω_M) to the total beam area (Ω_A).

$$BE\ (\varepsilon_M) = \frac{\Omega_M}{\Omega_A} = \frac{\text{Main beam area}}{\text{Total beam area}}$$

Total beam area (Ω_A) = Main beam area (Ω_M) + Minor lobe area (Ω_m)

Rack your Brain

What is a Whip antenna?

Previous Year's Questions

The radiation pattern of an antenna in spherical coordinates is given by-

$$F(\theta) = \cos^4\theta;\ \ 0 \le \theta \le \pi/2$$

The directivity of the antenna is-

(A) 10 dB (B) 12.6 dB
(C) 11.5 dB (D) 18 dB

i.e., $\Omega_A = \Omega_M + \Omega_m$

Divide above equation by Ω_A -

$$1 = \frac{\Omega_M}{\Omega_A} + \frac{\Omega_m}{\Omega_A}$$

$$1 + \varepsilon_M = \varepsilon_m$$

Since we know that-

$$\text{Stray Factor}\,(\varepsilon_m) = \frac{\text{Minor lobe area}}{\text{Total beam area}}$$

$$\therefore \varepsilon_m = \frac{\Omega_m}{\Omega_A}$$

Therefore, the main beam efficiency, $\varepsilon_M = \frac{\Omega_M}{\Omega_A}$

Antenna Beam Area (BA) or Beam Solid Angle (Ω_A)

- Antenna Beam Area is an area (ds) of the surface of the sphere seen from the center of sphere that subtends solid angle Ω_A.
- The total solid angle subtended by the sphere is 4π steradians (Sr)

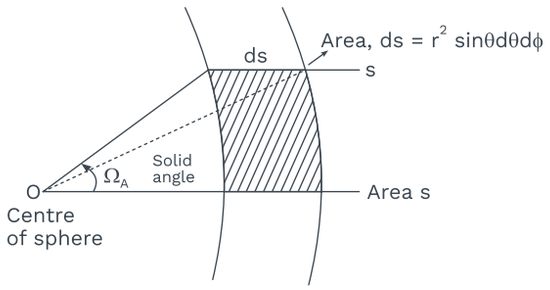


Figure 8.10: Solid Angle Ω_A subtended by Area s

As we know that-

ds = (r sin θ dϕ) (rdθ)

Rack your Brain

The signal level is 1 mW, the power gain will be _____ (dB_m).

Rack your Brain

The physical length of the Marconi antenna is _____.

$= r^2 \sin\theta \, d\theta \, d\phi$

$ds = r^2 \cdot d\Omega_A \, m^2$

where, $d\Omega_A$ = solid angle subtended by area ds

$$d\Omega_A = \frac{ds}{r^2} \quad sr$$

Thus, we get-

$$d\Omega_A = \frac{4\pi r^2}{r^2}$$

$d\Omega_A = 4\pi$

Also, we know that-

$$1 \text{ steradian} = \frac{d\Omega_A}{4\pi} = \frac{\text{Solid angle of sphere}}{4\pi}$$

$$1 \text{ sr} = 1(\text{radian})^2 = \left(\frac{180^\circ}{\pi}\right)^2 (\text{degrees})^2$$

$$1 \text{ sr} = \frac{180 \times 180}{3.14 \times 3.14} (\text{Degrees})^2$$

$$1 \text{ sr} = \frac{32400}{3.14 \times 3.14} (\text{Degrees})^2$$

$1 \text{ sr} = 3282.8 (\text{Degrees})^2$

Rack your Brain

The effective area of a Hertzian dipole operating at 100 MHz is 1.07 m^2

- Beam Area (or beam Solid Angle) Ω_A for the antenna is given by integral of normalized power pattern over a sphere (4π sr)-

$$\Omega_A = \int_0^{2\pi} \int_0^{\pi} P_n(\theta, \phi) d\Omega \quad Sr$$

$$\Omega_A = \int_0^{2\pi} \int_0^{\pi} P_n(\theta, \phi) \sin\theta d\theta \, d\phi \quad Sr$$

where $P_n(\theta, \phi)$ = Normalized Power Pattern

$$P_n(\theta, \phi) = \frac{P(\theta, \phi)}{P(\theta, \phi)_{max}}$$

- Solid Angle is described in terms of angles subtended by Half-power-points of main lobe in principal planes as-

  $\Omega_A = \theta_E \times \theta_H$ (Sr)

  θ_E = HPBW in E-plane or θ plane

  θ_H = HPBW in H-plane or ϕ plane

Effective aperture area

- It is defined as the ratio of power received to the time average poynting vector of the incident wave.

$$A_e = \frac{P_r}{(P_{avg})_{incident}}$$

- Relation between A_e and power gain is given as-

$$A_e = \frac{\lambda^2}{4\pi} G_P = \frac{\lambda^2}{4\pi} e_{cd} \left(1 - |\Gamma|^2\right) G_D$$

Rack your Brain

The directional pattern of a short dipole in the horizontal plane is a circle. (True/ False)

ANTENNA ARRAY

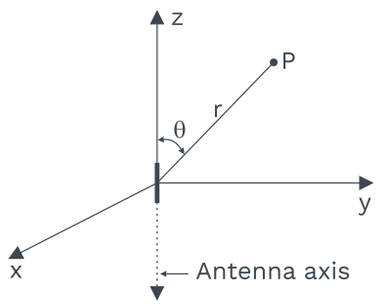


Figure 8.11: Antenna array

- Generally, we use an array of antenna elements in order to maximize the directivity.
- Consider a single dipole placed along the z-axis and at the origin.

- Antenna array is the group of radiating elements designed in order to achieve the desired directivity in a particular direction only.
- In the figure, consider a point P at distance 'r' and making an angle 'θ' with the z-axis. As dl<<<r, Thus the condition is of far-field.
- At point P, the radiation field are given as-

$$E_{\theta s} = \frac{\eta I_0\, dl \sin\theta\, e^{-j\beta r}(j\beta)}{4\pi r}$$

and $H_{\phi s} = \frac{E_{\theta s}}{\eta}$

Also, $E_{rs} = E_{\phi s} = H_{rs} = H_{\theta s} = 0$

- Considering the case of antenna array where the antenna element is placed horizontally unlike the prior case and is parallel to the y-axis.

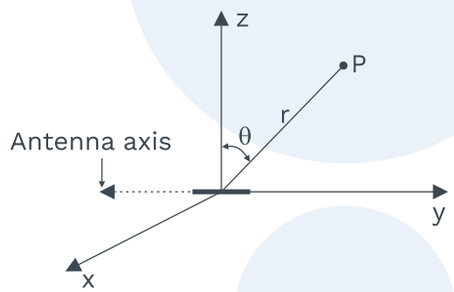


Figure 8.12

- In the above considered case at far-field, the radiation field will be-

$$E_{\theta s} = \frac{\eta I_0\, dl \sin(\theta + 90°)\, e^{-j\beta r}(j\beta)}{4\pi r} \text{ and } H_{\phi s} = \frac{E_{\theta s}}{\eta}$$

Also, $E_{rs} = E_{\phi s} = H_{rs} = H_{\theta s} = 0$

$$E_{\theta s} = \frac{\eta I_0\, dl \cos\theta\, e^{-j\beta r}(j\beta)}{4\pi r}$$

- Radiations from single dipoles like Hertzian dipole or Half-wave dipole are not sufficient to cover the desired directions.
  Thus, the idea of using multiple radiating elements is designed and called as an antenna array.
- This technique of using multiple elements is required to achieve the radiations in a desired direction or to get the desired radiation characteristics.

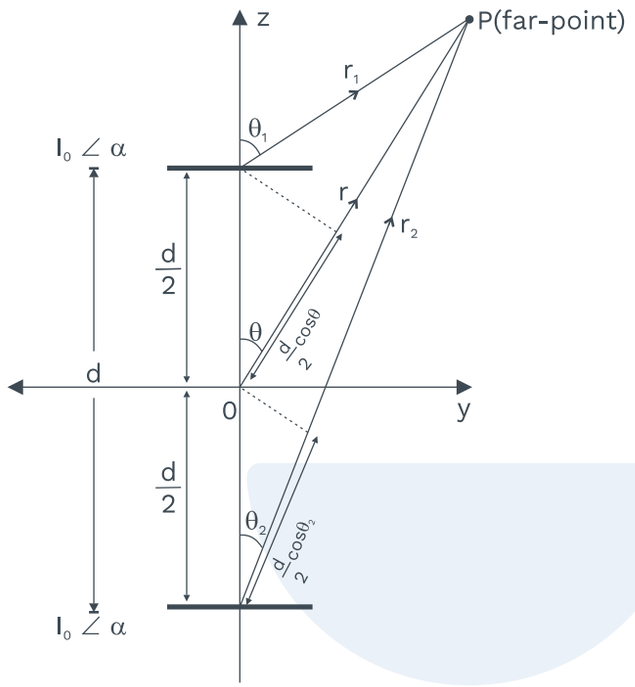


Figure 8.13: Two antenna element array

- Now, considering two antenna elements placed horizontally as shown in the figure above.
- The point 'P' is very far (dl <<< r_1) and (dl <<< r_2) or (r >>> d).
- Let for excitation of an antenna element-1, I_0 current is required with a phase shift of $\angle \alpha°$ and for antenna element-2, the current will be the same I_0 but the phase shift will be $\angle 0°$.
- Considering θ_1 and θ_2 angles are made by the given antenna elements with respect to the z-axis that are placed at 'r_1' and 'r_2' distance from point 'P'.
- The distance between two antenna elements is assumed as 'd'.
- At point 'P' , θ_1 θ θ_2 or we can say that the lines joining will [illegible]ome parallel to the z-axis.

$$r_1 = r - \frac{d}{2}\cos\theta$$

Similarly, $r_2 = r + \frac{d}{2}\cos\theta_2$

$(\because \theta = \theta_2)$

$$\therefore r_2 = r + \frac{d}{2}\cos\theta$$

These distance r_1 or r_2 doesn't affect the amplitude but have a significant change in the phase.

It can be said that there is negligible change in amplitude of radiation because of distance (r_1, r or r_2).

But there is a significant change in the phase of radiation due to 'r_1' and 'r_2'.

- For amplitude, $r_1 \simeq r \simeq r_2$ is required but in the case of phase shift, 'r_1' and 'r_2' are defined, as it is in the above equations.

$$E_{\theta s} = \frac{\eta I_0 \, dl \cos\theta \, e^{-j\beta r}(j\beta)}{4\pi r}$$

Here, $e^{-j\beta r}$ represents a phase and the remaining terms in the above equation represents the amplitude part.

- The sum of electric fields / radiations at point 'P' due to antenna element-1 and antenna element-2 will be a reference to as total radiation.

Mathematically-

$$E_{net} = E_1 + E_2$$

$$E_{net} = \frac{\eta I_0 \, dl \cos\theta_1 \, e^{-j\beta r_1}}{4\pi r_1} j\beta e^{j\alpha} + \frac{\eta I_0 \, dl \cos\theta_2 \, e^{-j\beta r_2}(j\beta)}{4\pi r_2}$$

$$E_{net} = \frac{\eta I_0 \, dl \cos\theta \, e^{-j\beta\left(r-\frac{d}{2}\cos\theta\right)} j\beta e^{j\alpha}}{4\pi r} + \frac{\eta I_0 \, dl \cos\theta \, e^{-j\beta\left(r+\frac{d}{2}\cos\theta\right)}(j\beta)}{4\pi r}$$

$\{\because \quad \theta_1 = \theta = \theta_2$ and $r_1 = r = r_2$ for amplitude part only$\}$

$$E_{net} = \frac{\eta I_0 \, dl \cos\theta (j\beta) \, e^{-j\beta_r}}{4\pi r}\left(e^{j\alpha} e^{j\beta\frac{d}{2}\cos\theta} + e^{-j\beta\frac{d}{2}\cos\theta}\right)$$

$$E_{net} = \frac{\eta I_0 \, dl \cos\theta \, e^{-j\beta_r}(j\beta)}{4\pi r} e^{j\frac{\alpha}{2}}\left(e^{j\frac{\alpha}{2}} e^{j\beta\frac{d}{2}\cos\theta} + e^{-j\frac{\alpha}{2}} e^{-j\beta\frac{d}{2}\cos\theta}\right)$$

$$E_{net} = \frac{\eta I_0 \, dl \cos\theta \, e^{-j\beta_r}}{4\pi r}(j\beta) e^{j\frac{\alpha}{2}}\left[e^{j\left(\frac{\alpha+\beta d\cos\theta}{2}\right)} + e^{-j\left(\frac{\alpha+\beta d\cos\theta}{2}\right)}\right]$$

$$E_{net} = \frac{\eta I_0 \, dl \cos\theta \, e^{-j\beta_r}}{4\pi r}(j\beta) e^{j\frac{\alpha}{2}} 2\cos\left(\frac{\alpha+\beta d\cos\theta}{2}\right)$$

As we know that-

$$E_{\theta s} = \eta \frac{I_0\, dl \cos\theta\, e^{-j\beta_r}}{4\pi r} j\beta$$

It is applicable when the antenna is placed horizontally at the origin only.

Thus,

$$E_{net} = \underbrace{\frac{\eta I_0 dl \cos\theta e^{-j\beta_r}(j\beta)}{4\pi r}}_{\text{Radiation due to single Hertzian dipole placed at origin (parallel to y – axis)}} \underbrace{e^{j\frac{\alpha}{2}} 2\cos\left(\frac{\alpha + \beta d\cos\theta}{2}\right)}_{\text{Array factor}}$$

- Resultant electric field = Field due to single dipole antenna placed at origin multiplied with array factor.
  Resultant electric field = Field due to single dipole antenna × Array factor.
- Now to draw the radiation pattern of the electric field, we determine the normalized value.
  Therefore, the normalized value of E_{net} will give the resultant radiation pattern for the antenna array with two elements.

$$\begin{bmatrix}\text{Resultant radiation}\\ \text{pattern of an array}\end{bmatrix} = \begin{bmatrix}\text{Radiation pattern due to single}\\ \text{antenna placed at origin}\end{bmatrix} \times \begin{bmatrix}\text{Pattern due to}\\ \text{array factor}\end{bmatrix}$$

  This above equation defines a law called pattern multiplication.
- The radiation pattern due to a single antenna place at origin is referred to as the unit pattern.
  And the pattern due to the array factor is referred to as a group pattern.
  Thus, the pattern multiplication can also be described as-
  Resultant radiation pattern = Unit pattern × Group pattern
- For unit pattern (E_1)

$$E_1 = \frac{\eta I_0\, dl \cos\theta e^{-j\beta r} j\beta}{4\pi r}$$

$$|E_1| = \frac{\eta I_0\, dl \cos\theta\, \beta}{4\pi r}$$

$$|E_1|_{max} = \frac{\eta I_0\, dl\, \beta}{4\pi r}$$

$$E_{norm} = \frac{|E_1|}{|E_1|_{max}} = \cos\theta$$

The normalized value will be-

$|E_{norm}| = |\cos\theta|$

Now plotting the radiation pattern

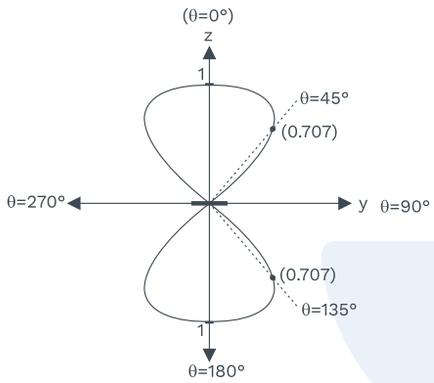


Figure 8.14: Unit Pattern

- **Group Pattern-**

Case I: When $\alpha = 0$ and $d = \frac{\alpha}{2}$

Array factor, $AF = e^{-j\frac{\alpha}{2}} 2\cos\left(\frac{\alpha + \beta d\cos\alpha}{2}\right)$

For normalization, we determine-

$$|AF| = 2\cos\left(\frac{\alpha + \beta d\cos\theta}{2}\right)$$

And $|AF|_{max} = 2$

Thus, $|AF|_{norm} = \frac{|AF|}{|AF|_{max}} = \cos\left(\frac{\alpha + \beta d\cos\theta}{2}\right)$

Substituting values of $\alpha = 0$ & $d = \frac{\alpha}{2}$

$$|AF|_{norm} = \cos\left(\frac{0 + \frac{2\pi}{\lambda} \times \frac{\lambda}{2}\cos\theta}{2}\right)$$

Previous Year's Questions

The far-zone power density radiated by a helical antenna is approximated as-

$$\vec{W}_{rad} = \vec{W}_{average} \simeq \hat{a}_r C_0 \frac{1}{r^2}\cos^4\theta$$

The radiated power density is symmetrical with respect to ϕ and exists only in the upper hemisphere: $0 \le \theta \le \frac{\pi}{2}$; $0 \le \phi \le 2\pi$; C_0 is a constant. The power radiated by the antenna (in watts) and the maximum directivity of the antenna, respectively, are-

(A) 1.5 C_0, 10 dB (B) 1.256 C_0, 10 dB
(C) 1.256 C_0, 12 dB (D) 1.5 C_0, 12 dB

Rack your Brain

The Effective area of and isotropic antenna is_____.

$$|AF|_{norm} = \cos\left(\frac{\pi}{2}\cos\theta\right)$$

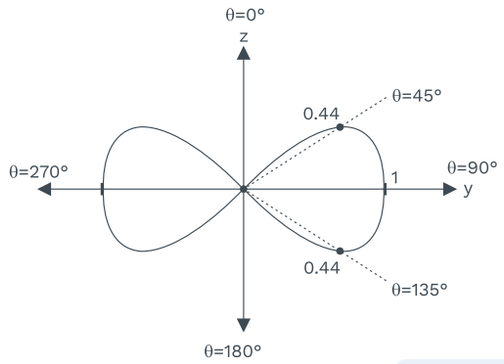


Figure 8.15: Group Pattern

θ	$\|AF\|_{norm}$
0	0
45°	0.44
90°	1
135°	0.44
180°	1

Table 8.1: Normalized array factor

- For resultant pattern-

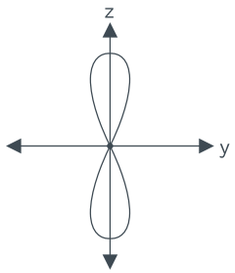


Figure 8.16(a): Unit Pattern

Previous Year's Questions

For an antenna radiating in free space, the electric field at a distance of 1 km is found to be 12 m V/m. Given that the intrinsic impedance of the free space is 120π Ω, the magnitude of average power density due to this antenna at a distance of 2 km from the antenna (in nW/m^2) is _____.

Rack your Brain

The effective area of an infinitesimal dipole is_____.

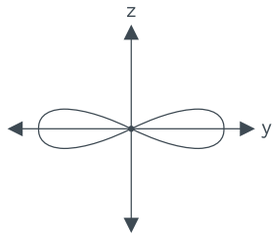


Figure 8.16(b): Group Pattern

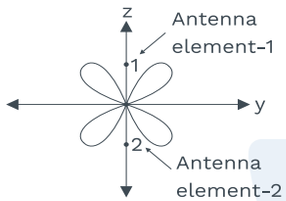


Figure 8.16(c): Resultant Pattern

- This radiation pattern is made in an E-plane pattern.

Note: The H-plane consists of x-y plane.

Case 2: When $\alpha = \frac{d}{2}$ and $d = \frac{\lambda}{4}$

$$|AF|_{norm} = \cos\left(\frac{\alpha + \beta d \cos\theta}{2}\right)$$

Substituting values-

$$|AF|_{norm} = \cos\left(\frac{\frac{\pi}{2} + \frac{2\pi}{\lambda} \times \frac{\lambda}{4}\cos\theta}{2}\right)$$

$$|AF|_{norm} = \cos\left(\frac{\pi}{4} + \frac{\pi}{4}\cos\theta\right)$$

$$|AF|_{norm} = \cos(45° + 45° \cos\theta)$$

Rack your Brain

The gain of infinitesimal dipole is_____.

Previous Year's Questions

A half-wave dipole is kept in the x-y plane and oriented along 45° from the x-axis. Determine the direction of null in the radiation pattern for $0 \le \phi \le \pi$. Here the angle $\theta(0 \le \theta \le \pi)$ is measured from the z-axis, and the angle $\phi(0 \le \phi \le 2\pi)$ is measured from the x-axis in the x-y plane.

(A) $\theta = 90°, \phi = 45°$
(B) $\theta = 45°, \phi = 90°$
(C) $\theta = 90°, \phi = 135°$
(D) $\theta = 45°, \phi = 135°$

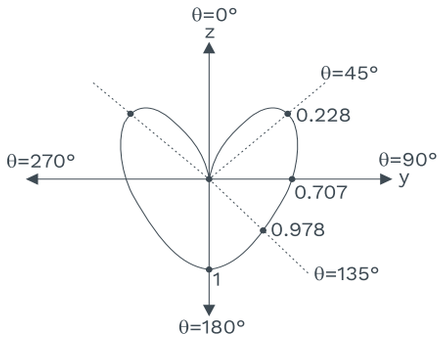


Figure 8.17: Group Pattern

θ	$\|AF\|_{norm}$
0	0
45°	0.228
90°	0.707
135°	0.978
180°	1

Table 8.2: Normalized array factor

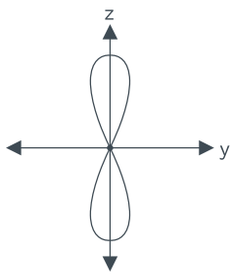


Figure 8.18(a): Unit Pattern

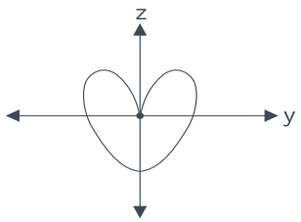


Figure 8.18(b): Group Pattern

Previous Year's Questions

Match column A with column B

Column A

1. Point electromagnetic source
2. Dish antenna
3. Yagi-Uda antenna

Column B

P. Highly directional
Q. End fire
R. Isotropic

(A) 1→R 2→Q 3→P
(B) 1→Q 2→P 3→R
(C) 1→R 2→P 3→Q
(D) 1→P 2→Q 3→R

Rack your Brain

The effective area of optimum horn whose mouth area is A_e is_____.

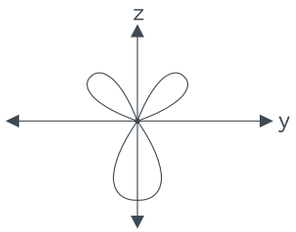


Figure 8.18(c): Resultant Pattern

Uniform Linear Array

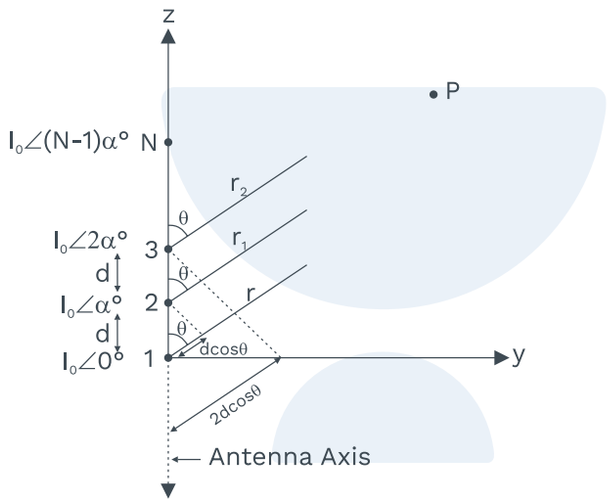


Figure 8.19

- Consider N-number of antenna elements placed along the z-axis at 'd' distance apart from each other and making θ angle with the z-axis. The point 'P' is at a far distance with respect to each antenna element.
- The excitation of each element is having phase shift as $I_0 \angle 0°$, $I_0 \angle \alpha$, $I_0 \angle 2\alpha$ ..... $I_0 \angle (N-1)\ \alpha$ for element 1, 2, 3 ..... N, respectively.
- According to the earlier derivation we calculate net electric field combining all respective electric fields due to each element.

  $E_{net} = E_1 + E_2 + E_3$, ... E_N

  $r_1 = r - d\cos\theta$, $r_2 = r - 2d\cos\theta$

Mathematically, we describe it as-

$$E_{net} = \frac{\eta I_0 dl\, e^{-j\beta r} j\beta \cos\theta}{4\pi r} + \frac{\eta I_0 dl \cos\theta\, e^{-j\beta(r-d\cos\theta)} e^{j\alpha}(j\beta)}{4\pi r}$$

$$+ \frac{\eta I_0 dl \cos\theta\, e^{2\alpha} e^{-j\beta(r-2\cos\theta)}(j\beta)}{4\pi r} \ldots\ldots \frac{\eta I_0 dl e^{-j(N-1)\alpha} \sin\theta\, e^{-j\beta[r-(N-1)d\cos\theta]}(j\beta)}{4\pi r}$$

$$E_{net} = \frac{\eta I_0 dl \cos\theta\, e^{-j\beta r}(j\beta)}{4\pi r}[1 + e^{j\alpha} e^{j\beta d\cos\theta} + e^{j2\alpha} e^{j\beta 2d\cos\theta} \ldots\ldots\ldots\ldots + e^{j(N-1)\alpha} e^{j\beta(N-1)d\cos\theta}]$$

$$E_{net} = \frac{\eta I_0 dl \cos\theta\, e^{-j\beta r}(j\beta)}{4\pi r}[1 + e^{j(\alpha+\beta d\cos\theta)} + e^{j2(\alpha+\beta d\cos\theta)} \ldots\ldots\ldots\ldots + e^{j(N-1)(\alpha+\beta d\cos\theta)}]$$

Let, $E_0 = \frac{\eta I_0 dl \sin\theta\, e^{-j\beta r}(j\beta)}{4\pi r}$

and $\psi = \alpha + \beta d\cos\theta$

$\therefore \quad E_{net} = E_0[1 + e^{j\psi} + e^{j2\psi} + \ldots\ldots + e^{j(N-1)\psi}]$

$$E_{net} = \frac{E_0(1 - e^{j\psi N})}{1 - e^{j\psi}}$$

$$E_{net} = E_0 \frac{e^{j\frac{N\psi}{2}}\left[e^{-j\frac{N\psi}{2}} - e^{j\frac{N\psi}{2}}\right]}{e^{j\frac{\psi}{2}}\left[e^{-j\frac{\psi}{2}} - e^{j\frac{\psi}{2}}\right]}$$

Previous Year's Questions

An antenna with a directive gain of 6 dB is radiating a total power of 16 kW. The amplitude of electric field in free space at a distance of 8 km from the antenna in the direction of 6 dB gain is (Round off to 3 decimal places) _____V/m.

Now, dividing by 2j in numerator and denominator, we get-

$$E_{net} = E_0\, e^{j(N-1)\frac{\psi}{2}}\left[\frac{\sin\left(\frac{N\psi}{2}\right)}{\sin\left(\frac{\psi}{2}\right)}\right]$$

where E_0 = electric field due to single dipole/ Hertzian dipole placed at the origin.

$$\text{Phase} = e^{j(N-1)\frac{\psi}{2}}$$

$$\text{Array factor, } |AF| = \frac{\sin\left(\frac{N\psi}{2}\right)}{\sin\left(\frac{\psi}{2}\right)}$$

(i) For maximum radiation (or principal beam)

Put $\psi = 0$

$\therefore \alpha + \beta d \cos\theta_m = 0$

where-

θ_m = Angle at which we get maximum radiation

$$\beta d \cos\theta_m = -\alpha$$

$$\cos\theta_m = \frac{-\alpha}{\beta d}$$

$$\theta_m = \cos^{-1}\left[\frac{-\alpha}{\beta d}\right]$$

$$\theta_m = \cos^{-1}\left[\frac{\alpha}{\beta d}\right]$$

$\therefore$ Array factor (AF) is expressed as-

$$|AF| = \frac{\sin\left(\frac{N\psi}{2}\right)}{\sin\left(\frac{\psi}{2}\right)} = \frac{0}{0}$$

Thus, by L'Hospital Rule, we get-

$$|AF|_{max} = \lim_{\psi \to 0} \frac{\frac{N}{2}\cos\frac{N\psi}{2}}{\frac{1}{2}\cos\frac{\psi}{2}}, \quad |AF|_{max} = N$$

and $$|AF|_{norm} = \frac{|AF|}{|AF|_{max}} = \frac{\sin\left(\frac{N\psi}{2}\right)}{N\sin\left(\frac{\psi}{2}\right)}$$

where, N = Number of radiating elements.

(ii) Null will occur, when |AF| = 0.

i.e., $$\sin\left(\frac{N\psi}{2}\right) = \sin(m\pi)$$

$$\sin\left(\frac{N\psi}{2}\right) = 0$$

Previous Year's Questions

The radiation resistance of a small dipole current element of length l at a frequency of 3 GHz is 3 ohms. If the length is changed by 1%. Then the percentage change in the radiation resistance rounded off to two decimal places, is _____%

Rack your Brain

The directional characteristics of center-fed 2λ dipole is_____.

$$\frac{N\psi}{2} = \pm m\pi$$

$$\frac{N}{2}(\alpha + \beta d\cos\theta_N) = \pm m\pi$$

θ_N = Angle at which null will occur.

If ψ <<< 1,

Then, $\sin\frac{\psi}{2} \approx \frac{\psi}{2}$

$$|AF|_{norm} = \frac{\sin\left(\frac{N\psi}{2}\right)}{\frac{N\psi}{2}}$$

- For two elements in an array-

  N = 2

$$|AF|_{norm} = \frac{\sin\psi}{2\sin\psi} = \frac{2\sin\frac{\psi}{2}\cos\frac{\psi}{2}}{2\sin\frac{\psi}{2}}$$

and also, we know that-

$$|AF|_{norm} = \cos\frac{\psi}{2} = \cos\left(\frac{\alpha + \beta d\cos\theta}{2}\right)$$

(iii) **For Broadside Array:**

It is the array in which maximum radiation occurs in the direction perpendicular to the antenna axis (z-axis).

Then,

ψ = 0 and θ_m = 90°

$\therefore \alpha + \beta\cos\theta_m = 0$

α = 0 (progressive phase shift)

(iv) **End-fire array:**

It is an array where maximum radiation occurs in direction of the array axis (z-axis).

Maximum radiation occurs at-

ψ = 0

$\therefore \alpha + \beta d\cos\theta$

Rack your Brain

For the broadside array, the excitation phase should be _____.

Previous Year's Questions

An array of 10 isotropic elements is placed along the z-axis at distance $\frac{\lambda}{4}$.

Assume the uniform distribution for a broad-side array.

Calculate beamwidth first Null (BWFN) (in degree) and directivity (in dB).

(A) 45.83; 6.98 (B) 45; 6

(C) 46.98; 10 (D) 45.13; 20

Rack your Brain

ϕ-polarization is synonymous with horizontal polarization.

$$\alpha = -\beta d \cos\theta$$

$$\text{if } \begin{cases} \theta_m = 0 & ; \quad \alpha = -\beta d \\ \theta_m = 180° & ; \quad \alpha = \beta d \end{cases}$$

$$\therefore \alpha = \pm \beta d$$

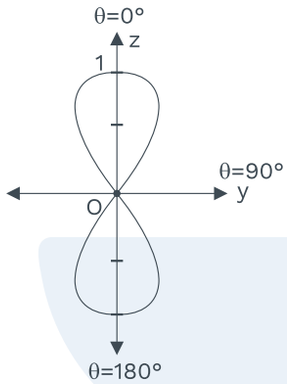


Figure 8.20

- Calculation of HPBW and BWFN for antenna arrays-

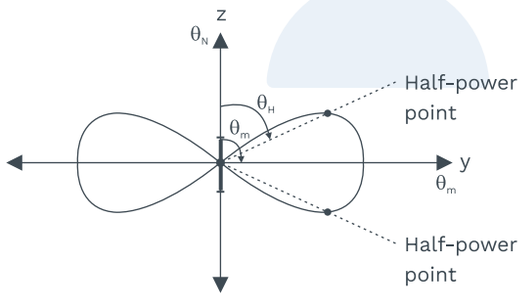


Figure 8.21

As we know that Half power beam width is-

HPBW = $2|\theta_m - \theta_H|$

and Beamwidth between first null is-

BWFN = $2|\theta_m - \theta_N|$

For Hertzian dipole-

$\theta_m = 90°$ and $\theta_N = 0$

$\therefore$ BWFN = 2(90°) = 180° and HPBW = 90°

where θ_N = Angle where main beam radiation is zero.

Rack your Brain

For the end-fire array, the progressive phase shift should be equal to

(i) Broadside array ($\alpha = 0$):-
As we know that-

$$\theta_m = \cos^{-1}\left(\frac{\alpha}{\beta d}\right)$$

θ_H = Angle at which array factor $|AF|_{norm}$ becomes 0.707

$|AF|_{norm} = 0.707$

or $$\frac{N\psi}{2} = \pm 1.391$$

$N(\alpha + \beta d\cos\theta_H) = \pm 2.782$

$$\beta d\cos\theta_H = \pm\frac{2.782}{N} - \alpha$$

$$\cos\theta_H = \frac{-\alpha}{\beta d} \pm \frac{2.782}{\beta dN}$$

$$\theta_H = \cos^{-1}\left[\frac{-\alpha\lambda}{2\pi d} \pm \frac{2.782\lambda}{2\pi dN}\right] \quad \left[\because \beta = \frac{2\pi}{\lambda}\right]$$

- Now, HPBW is given as-
  $HPBW = 2|\theta_m - \theta_H|$

$$HPBW = \left|\cos^{-1}\left(\frac{\alpha}{\beta d}\right) - \cos^{-1}\left(\frac{-\alpha\lambda}{2\pi d} \pm \frac{2.782\lambda}{2\pi Nd}\right)\right|$$

- For broad side array ($\alpha = 0$)

$$HPBW = 2\left|\frac{\pi}{2} - \cos^{-1}\left(\pm\frac{2.782\lambda}{2\pi Nd}\right)\right|$$

$$HPBW = 2\left|\sin^{-1}\left(\pm\frac{2.782}{2\pi Nd}\right)\right| = 2\left|\sin^{-1}\left(\frac{0.44\lambda}{Nd}\right)\right|$$

- For a large value of $d >>> \lambda$

$$HPBW = 2\left(\frac{0.44\lambda}{Nd}\right)$$

$$HPBW = \frac{0.88\lambda}{Nd}$$

where, N = Number of radiating.

Rack your Brain

The effect of the earth on vertical patterns is that the directivity increases and the radiation is in the upper hemisphere. (True/False)

d = spacing between two elements.
λ = wavelength.

- Beamwidth between first null (BWFN) for broadside array will be-
As we know-

$$\frac{N}{2}(\alpha + \beta d\cos\theta_N) = \pm m\pi$$

$$\alpha + \beta d\cos\theta_N = \pm\frac{2m\pi}{N}$$

$$\cos\theta_N = \pm\frac{2m\pi}{N\beta d} - \frac{\alpha}{\beta d}$$

$$\theta_N = \cos^{-1}\left[-\frac{\alpha}{\beta d} \pm \frac{2m\pi}{N\beta d}\right]$$

Thus, we know-

$$BWFN = 2|\theta_M - \theta_N|$$

and $\theta_m = \cos^{-1}\left(\frac{\lambda}{\beta d}\right)$

Substituting the values in BWFN we get-

$$BWFN = 2\left[\cos^{-1}\left(\frac{\lambda}{\beta d}\right) - \cos^{-1}\left(-\frac{\alpha}{\beta d} \pm \frac{2m\pi}{N\beta d}\right)\right]$$

where, m = integer

Rack your Brain

Does an ungrounded antenna near the ground act as an antenna array?

- For broadside array ($\alpha = 0$)

$$\therefore BWFN = 2\left|\frac{\pi}{2} - \cos^{-1}\left(\frac{2m\pi\lambda}{N2\pi d}\right)\right|$$

$$BWFN = 2\sin^{-1}\left[\pm\frac{m\lambda}{Nd}\right]$$

We get first null at m = 1 and d >>> λ

$$\therefore BWFN = \frac{2\lambda}{Nd}$$

Rack your Brain

The sidelobe level of a binomial array is_____.

- Directivity for broadside array will be-

$$\text{Directivity} = \frac{2Nd}{\lambda}$$

Parameter	Broadside	Endfire array
HPBW	$\frac{0.88\lambda}{Nd}$	$2\sqrt{\frac{0.88\lambda}{Nd}}$
BWFN	$\frac{2\lambda}{Nd}$	$2\sqrt{\frac{2\lambda}{Nd}}$
Directivity	$\frac{2Nd}{\lambda}$	$\frac{4Nd}{\lambda}$

Table 8.3: Formulas for Broadside and Endfire Antennas

Example 10:

In a free space, an antenna has a far-zone field given by $\vec{E}_s = \frac{5\sin 2\theta}{r} e^{-j\beta r}\,\hat{a}_r$ V/m.

Find the radiated power.

Solution:

The electric field for an antenna in free space is given by-

$$\vec{E}_s = \frac{5\sin 2\theta}{r} e^{-j\beta r}\,\hat{a}_r \text{ V/m}$$

And the radiated power is determined by-

$$P_{rad} = \int_s \vec{P}_{avg} \cdot d\vec{s}$$

$$\vec{P}_{avg} = \frac{E_0^2}{2\eta}\hat{a}_r \text{ and } d\vec{s} = \hat{a}_r r^2 \sin\theta d\phi$$

$$\vec{P}_{avg} = \left(\frac{5\sin 2\theta}{r}\right)^2 \frac{1}{2\eta}\hat{a}_r = \frac{25\sin^2 2\theta}{r^2}\hat{a}_r$$

Since, $(\sin 2\theta = 2\sin\theta\cos\theta)$

$$\therefore \vec{P}_{avg} = \frac{25(2\sin\theta\cos\theta)^2}{r^2(2\eta)} = \frac{100\sin^2\theta\cos^2\theta}{r^2(2\eta)}\hat{a}_r = \frac{50\sin^2\theta\cos^2\theta}{r^2\eta}$$

Rack your Brain

The beamwidth of a binomial array is greater than that of a uniform linear array. (True/False)

Now,

$$P_{rad} = \int_s \vec{P}_{avg} \cdot d\vec{s}$$

$$P_{rad} = \int_s \frac{50 \sin^2\theta \cos^2\theta}{\eta r^2} \hat{a}_r \cdot \hat{a}_r \, r^2 \sin\theta d\theta d\phi$$

$$P_{rad} = \int_s \frac{50}{120\pi} \sin^3\theta \cos^2\theta d\theta d\phi \qquad \{\text{For free space, } \eta = 120\pi\}$$

$$P_{rad} = \frac{50}{120\pi} \int_{\theta=0}^{\pi} \sin^3\theta \cos^2\theta d\theta \int_{\phi=0}^{2\pi} d\phi$$

$$P_{rad} = \frac{100}{120\pi} \int_0^{\pi/2} \sin^3\theta \cos^2\theta d\theta \int_{\phi=0}^{2\pi} d\phi = \frac{5}{6\pi}\left(\frac{2\times 1}{5\times 3\times 1}\right) \times 1 \times 2\pi$$

$$P_{rad} = \frac{2}{9} = 222.2\,mW$$

Since, this property we get-

$$\int_0^{\pi/2} \sin^m\theta \cos^n\theta \, d\theta = \frac{[(m-1)(m-3)\ldots(2 \text{ or } 1)][(n-1)(n-3)\ldots(2 \text{ or } 1)]}{[(m+n)(m+n-2)\ldots(m+n-4)\ldots(2 \text{ or } 1)]}$$

where $\begin{cases} K = \frac{\pi}{2} & ; \text{ if both m \& n are even} \\ K = 1 & ; \text{ otherwise} \end{cases}$

The above formula is applicable for-

$$0 < \theta < \frac{\pi}{2}$$

Example 11:
At the far-field, an antenna produces

$$\vec{P}_{avg} = \frac{2\sin\theta\cos\phi}{r^2} \hat{a}_r \, W/m^2$$

where, $0 < \theta < \pi$ and $0 < \phi < \frac{\pi}{2}$

Evaluate the directive gain and directivity of the antenna.

Rack your Brain

The radiation resistance of a dipole due to the earth effect is _____.

Solution:

Given, the time average pointing vector is-

$$\vec{P}_{avg} = \frac{2\sin\theta\cos\phi}{r^2}\hat{a}_r \text{ W/ m}^2$$

and the power radiated is given by-

$$P_{rad} = \int_s \vec{P}_{avg} \cdot d\vec{s}$$

$$P_{rad} = \int_{\theta=0}^{\pi}\int_{\phi=0}^{\pi/2} \frac{2\sin\theta\cos\phi}{r^2} r^2 \sin\theta d\theta d\phi =$$

$$\int_{\phi=0}^{\pi/2} \cos\phi d\phi \int_{\theta=0}^{\pi} 2\sin^2\theta\, d\theta$$

$$P_{rad} = [+\sin\phi]_0^{\pi/2} \times \left[2 \times \frac{2\times(2-1)}{2} \times \frac{\pi}{2}\right]$$

$$\therefore P_{rad} = \pi$$

The directive gain will be-

$$G_D(\theta,\phi) = \frac{2\sin\theta\cos\phi \times 4\pi}{\pi}$$

$G_D(\theta, \phi) = 8\sin\theta\cos\phi$

Thus, the directivity will be-

$D = \max[G_D(\theta, \phi)]$

Directivity = 8

Example 12:

A microwave antenna has a power gain of 15dB at 9 GHz. Determine its effective area.

Solution:

Given that-

Power gain $(G_P)dB = 10\log G_P = 15dB$

$\therefore \quad 15 = 10\log G_p$

$G_P = 10^{1.5} = 31.62$

Gray Matter Alert!!!

The standard reference antenna for directive gain is the isotropic antenna.

Rack your Brain



θ polarization is synonymous with vertical polarization. (True/False)

Effective area is given by-

$$A_e = \frac{\lambda^2}{4\pi} G_P$$

Since, $f = 9 \times 10^9$ Hz

$$\therefore \lambda = \frac{c}{f} = \frac{3 \times 10^8}{9 \times 10^9} = \frac{3}{90} = \frac{1}{30}$$

$$A_e = \frac{\lambda^2}{4\pi} \times G_P = \left(\frac{1}{30}\right)^2 \times \frac{1}{4\pi} \times 31.62$$

$$A_e = \frac{1}{900} \times \frac{1}{4\pi} \times 31.62 = 2.75 \text{ m}$$

Example 13:

Calculate U_{avg}, U_{max} and D if-

(A) $U(\theta, \phi) = \sin^2\theta$ $\{0<\theta<\pi ; 0<\phi<2\pi\}$

(B) $U(\theta, \phi) = 2\sin^2\theta\sin^2\phi$ $\{0<\theta<\pi ; 0<\phi<2\pi\}$

Solution:

(A) Given radiation intensity –

$U(\theta, \phi) = \sin^2\theta$ $\{0<\theta<\pi ; 0<\phi<2\pi\}$

Maximum radiation intensity, $U_{max} = 1$

And $U_{avg} = \frac{P_{rad}}{4\pi}$

$$P_{rad} = \int_s U(\theta, \phi) \sin\theta d\theta d\phi$$

$$P_{rad} = \int_{\phi=0}^{2\pi} \int_{\phi=0}^{\pi} (2\sin\theta\cos\theta)^2 \sin\theta \, d\theta$$

$$P_{rad} = 2\pi \times 4 \int_0^{\pi} \sin^3\theta \cos^2\theta \, d\theta$$

$$P_{rad} = 16\pi \int_0^{\pi/2} \sin^3\theta \cos^2\theta \, d\theta$$

$$P_{rad} = 16\pi \frac{2 \times 1}{5 \times 3 \times 1}$$

Rack your Brain

The radiation patterns of terminated and non-terminated antennas are the not same. (True/False)

Gray Matter Alert!!!

The effective area of a Hertzian dipole operating at 100 MHz is 1.07 m^2

$\therefore P_{rad} = \frac{32\pi}{15}$

$\therefore$ The average value-

$U_{avg} = \frac{32\pi}{15 \times 4\pi} = \frac{8}{15} = 0.533$

and the directivity will be-

$D = \frac{4\pi U_{max}}{P_{rad}} = \frac{4\pi \times 1 \times 15}{32\pi}$

$\therefore D = \frac{15}{8} = 1.875$

(B) Given radiation intensity–

$U(\theta, \phi) = 2\sin^2\theta\sin^2\phi \qquad \{0<\theta<\pi \ ; \ 0<\phi<2\pi\}$

Here, Maximum radiation intensity is $U_{max} = 2$

and $P_{rad} = \int_s U(\theta, \phi)\sin\theta \, d\theta \, d\phi$

$= \int_s 2\sin^3\theta \sin^2\phi \ d\theta \, d\phi$

$P_{rad} = \int_{\phi=0}^{\pi} 2\sin^2\phi \, d\phi \int_{\theta=0}^{\pi} \sin^3\theta \, d\theta = 8\int_{\phi=0}^{\pi/2} \sin^2\phi \, d\phi \int_{\theta=0}^{\pi/2} \sin^3\theta \, d\theta$

$P_{rad} = 8\left(\frac{1}{2} \times \frac{\pi}{2}\right)\left(\frac{2}{3 \times 1}\right)$

$\therefore P_{rad} = \frac{4\pi}{3}$

and $U_{avg} = \frac{P_{rad}}{4\pi} = \frac{1}{3}$

Directivity, $D = \frac{U_{max}}{U_{avg}} = \frac{2}{1/3}$

$\therefore D = 6$

Rack your Brain

Basic transmission loss between two antennas depends on which of the following factors-
(A) frequency
(B) distance
(C) frequency and distance
(D) gain of antennas.

Rack your Brain

Actual transmission loss between two antennas depends on what factors?

Example 14:
An antenna radiates a field in the free space at far-field-

$$E_{\phi s} = \frac{0.4\cos^2\theta}{4\pi r} e^{-j\beta r} \text{ kV/ m}$$

Evaluate-

(A) Total radiated power

(B) The directive gain at $\theta = 60°$

Rack your Brain

Antenna temperature is the temperature of the antenna. (Yes/No)

Solution:
(A) The given far-field of an antenna is-

$$E_{\phi s} = \frac{0.4\cos^2\theta}{4\pi r} e^{-j\beta r} \text{ kV/ m}$$

The radiated power will be-

Rack your Brain

When the sidelobe level increases, beam width also increases in general. (True/False)

$$P_{rad} = \int_s \vec{P}_{avg} \cdot d\vec{s}$$

$$P_{rad} = \int_s \left(\frac{0.4\cos^2\theta}{4\pi r}\right)^2 \times \frac{1}{2\eta} \hat{a}_r \cdot \hat{a}_r \, r^2 \sin\theta d\theta d\phi$$

$$P_{rad} = \int_s \frac{0.16 \times 1 \times 10^6}{(4\pi)_r^2 (240\pi)} \times \int_{\phi=0}^{2\pi} d\phi \int_{\theta=0}^{\pi} \cos^4\theta \sin\theta \, d\theta$$

$$P_{rad} = \frac{0.16 \times 10^6}{16\pi^2 \times 240\pi} \times 2\pi \times \frac{2\pi \times 3 \times 1}{5 \times 3 \times 1} = \frac{16 \times 4\pi \times 10^6}{16\pi^3 \times 5 \times 240 \times 100} \quad \frac{4}{5\pi^2 240} \times 10^4$$

$\therefore P_{rad} = 3.37$ W

(B) The directive gain at $\theta = 60°$ will be-

$$G_D(\theta, \phi) = \frac{4\pi U(\theta, \phi)}{P_{rad}}$$

$$G_D(\theta, \phi) = \frac{4\pi r^2 \left(\frac{0.4\cos^2\theta}{4\pi r}\right)^2}{3.77 \times 240\pi} = \frac{4\pi \times 0.16(\cos^4 60°)}{(4\pi)^2 \times 3.37 \times 240\pi}$$

$G_D(\theta, \phi) = 3.313$

FRIIS TRANSMISSION EQUATION

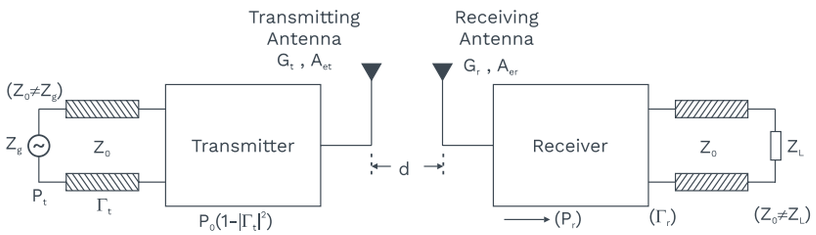


Figure 8.22: Friis Space Transmission

- Consider the distance between transmitting antenna and receiving antenna as 'd'.
  Let the gain of transmitting antenna as G_t and effective aperture area as A_{et}.
  Similarly, assume for receiving antenna, the power gain will be G_r and the effective aperture area will be A_{er}.
- Assuming, $Z_g \neq Z_0$ and power transmitting P_t are being generated and there will be reflection coefficient at transmitting side as Γ_t.

$\therefore$ The power transmitted will be-

$$P_t(1 - |\Gamma_t|)^2$$

- The power transmitted from the antenna is $P_t(1 - |\Gamma_t|^2).G_t$
- The power density at distance 'd' is-

$$= \frac{\text{Power}}{\text{Area}} = \frac{P_t(1-|\Gamma_t|^2)G_t}{4\pi d^2} \quad \frac{W}{m^2}$$

Since transmitting antenna is radiating a spherical symmetry.
Hence, the area of a sphere = $4\pi d^2$, where d is the distance.

- Now at receiving end, the area on which the transmitted signal is intercepted and absorbed will be referred to as the effective aperture area (A_{er}).
  And the power received at distance 'd' is-

$$P_r = \frac{P_t[1-|\Gamma_t|^2]G_t A_{er}}{4\pi d^2}$$

- Power delivered to the load will be-

$$P_L = \frac{P_t[1-|\Gamma_t|^2][1-|\Gamma_r|^2]G_t A_{er}}{4\pi d^2}$$

- We know that-
  Effective aperture area is-

$$A_e = \frac{\lambda^2}{4\pi} G$$

$$\text{Thus, } P_L = P_t G_t \left(\frac{\lambda^2 G_r}{4\pi}\right) \frac{\left[1-|\Gamma_t|^2\right]\left[1-|\Gamma_r|^2\right]}{4\pi d^2}$$

$$P_L = P_t G_t G_r \left(\frac{\lambda}{4\pi d}\right)^2 \left[1-|\Gamma_t|^2\right]\left[1-|\Gamma_r|^2\right]$$

- For perfectly matched condition: -

  $Z_g = Z_0$ and $Z_L = Z_0$

$$\Gamma_r = \frac{Z_L - Z_0}{Z_L + Z_0} = 0$$

$$\Gamma_t = \frac{Z_g - Z_0}{Z_g + Z_0} = 0$$

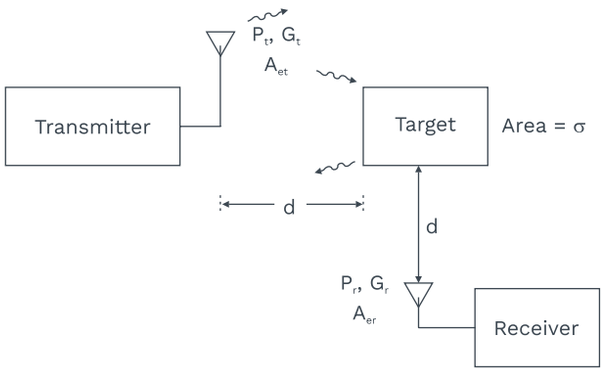
  Therefore, for perfectly matched conditions-

$$P_L = P_t G_t G_r \left(\frac{\lambda}{4\pi d}\right)^2$$

Rack your Brain

The relation between effective noise temperature and noise figure is _____.

Example 15:



Consider a radar system where the distance between the transmitting antenna and target is 'd' and the same distance 'd' is from the target and receiving antenna. Assume reflections efficiency is 0 percent here. Determine the power received by the transmitting antenna.

Solution:
Let the power transmitted = P_t
and distance antenna gain = G_t
The power received will be-

$$P_r = \frac{P_t G_t \sigma}{4\pi d^2} \times \frac{1}{4\pi d^2} \times A_{er}$$

$$P_t = \frac{P_t G_t A_{er} \sigma}{(4\pi d^2)^2}$$

$$P_t = \frac{P_t G_t \sigma}{(4\pi d^2)^2} \times \left(\frac{G_r \lambda^2}{4\pi}\right)$$ is the required power received.

Gray Matter Alert!!!

dBi means the power gain of the antenna in dB relative to an isotropic antenna.

Example 16:
The electric field strength incident on a Half-wave dipole is 4.6 m V/m at 50MHz. The maximum power received by the antenna will be _____.
[Assume the directivity of the Half-wave dipole as 1.64]

Solution:
The electric field strength of a given Half-wave dipole = 4.6m V/m.
Frequency = 50MHz
Gain = 1.64
The effective aperture area of an antenna is-

$$A_e = \frac{P_r}{(P_{avg})_{\text{incident wave}}}$$

$$P_r = A_e (P_{avg})_{\text{incident}} = \frac{\lambda^2}{4\pi} \times G \times \frac{E_0^2}{2\eta}$$

$$P_r = \left(\frac{3 \times 10^8}{50 \times 10^6}\right) \frac{1}{4\pi} \times 1.64 \times \frac{(4.6 \times 10^{-3})^2}{2 \times 120\pi}$$

$$P_r = \frac{36 \times 1}{4\pi} \times 1.64 \times \frac{21.16 \times 10^{-6}}{240\pi}$$

P_r = 131.85nW

Hence, the maximum power received by the antenna will be 131.85 nW.

Rack your Brain

The beamwidth between first null is _____ the half-power beamwidth.

Example 17:
Consider a synchronous orbit satellite antenna. The power transmitted by the satellite antenna is 425W. If the antenna has a gain of 45dB. Determine the power received by another antenna with a gain of 20dB at the range of 30,000km at a frequency of 15GHz.

Solution:
The power received will be-

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d}\right)^2$$

$$P_r = 425 \times 10^{4.5} \times 10^2 \left[\frac{3 \times 10^8}{15 \times 10^9 \times 4\pi \times 30,000 \times 10^3}\right]^2$$

$$P_r = 1343968005.58 \left[\frac{1}{3600\pi^2} \times 10^{-16}\right]$$

$$P_r = 3.782 \text{ pW}$$

Polarization Loss Factor (PLF)

- Generally, the polarization of the receiving antenna will not be the same as the polarization of the incoming (incident) wave.
  This phenomenon is generally referred to as 'polarization mismatch'.
- The amount of power extracted by the antenna from the incoming signal will be maximum because of the polarization losses.
- Assuming that the electric field of the incoming wave as-

  $\vec{E}_i = E_i \hat{\rho}_w$

  where, $\hat{\rho}_w$ is the unit vector of the wave.
- The polarization of the antenna and the polarization of the wave must be the same so as to get the maximum amount of power that can be extracted from the incident wave by the antenna.
- The polarization of the electric field of the receiving antenna is given by-

  $\vec{E}_a = E_a \hat{\rho}_a$

  here, $\hat{\rho}_a$ is the unit vector of the polarization vector $\vec{E}_a$.

- The polarization loss can be understood by considering a polarization loss factor (PLF).
- PLF can be defined as below based on the polarization of the antenna in its transmitting-

$$PLF = |\hat{\rho}_w \cdot \hat{\rho}_a|^2 = |\cos\psi_P|^2$$

- In other words, PLF is the square of the magnitude of the dot product of $\hat{\rho}_w$ (unit vector of $\vec{E}_i$) and $\hat{\rho}_a$ (unit vector of $\vec{E}_a$.
- PLF is a dimensionless quantity and ψ_p is defined as the angle between the unit vector $\hat{\rho}_w$ and $\hat{\rho}_a$.

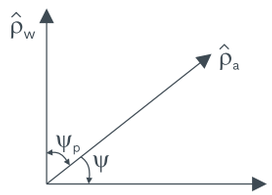


Figure 8.23

Example 18:
The directive gain of an antenna is 30dB. If the antenna radiates 6.45kW. Calculate the time-average power density at that distance.

Solution:
Given-
Directive gain $G_D(\theta, \phi) = 30dB = 10^3$
Power radiated = $6.45kW = 6.45 \times 10^3$ W
Distance, r = $45km. = 45 \times 10^3$ m
The time-average power density will be-

$$P_{avg} = \frac{G_D(\theta,\phi) \times P_{rad}}{4\pi r^2}$$

Rack your Brain

What are the disadvantages of side lobes?

Gray Matter Alert!!!

Arrays are generally used to increase the gain just like cascaded amplifiers and are also used to increase the gain. Arrays are also used to increase directivity and reduce beamwidth.

As we know, $G_D(\theta,\phi) = \frac{4\pi U(\theta,\phi)}{P_{rad}}$

$$G_D(\theta,\phi) = \frac{4\pi r^2 P_{avg}}{P_{rad}}$$

$$\therefore\ P_{avg} = \frac{G_D(\theta,\phi) \times P_{rad}}{4\pi r^2}$$

$$P_{avg} = \frac{10^3 \times 6.45 \times 10^3}{4\pi(45 \times 10^3)^2} = \frac{6.45 \times 10^6}{4\pi x\ \ x10^6}$$

$P_{avg} = 2.534 \times 10^{-4} = 0.253 \times 10^{-3}$ W/m^2

$P_{avg} = 0.253$ mW/m^2

Example 19:

Consider two identical antennas in an anechoic chamber that are separated by 10m and are oriented for maximum directive gain. The frequency is 6GHz and the power received by one is 40dB down from that transmitted by the other. Determine the gain of the antenna in dB.

Solution:

Given that-

The power received is 40dB down from transmitted power is expressed as-

$$P_r = \frac{P_t}{10000}$$

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d}\right)^2$$

$$\frac{P_t}{10000} = P_t G^2 \left(\frac{3 \times 10^8}{6 \times 10^9 \times 4\pi \times 10}\right)^2$$

Since, for identical antenna, $G_t = G_r = G$

$$\frac{P_t}{10000} = P_t G^2 \left(\frac{1}{800\pi}\right)^2$$

$$\frac{640000\pi^2}{10000} = G^2$$

Rack your Brain

The number of sources required in a uniform array of 10 elements is _____.

Gray Matter Alert!!!

When the transmitting antenna is excited by an alternating voltage, the initial motion is started by the balanced charges movements in the antenna. Resonant oscillations are produced by the supplied energy.

$64\pi^2 = G^2$

$G = 8\pi = 25.13$

Gain (in dB) = $10\log(25.13) \simeq 14$dB

Example 20:
Determine the maximum power that can be received over the 2km distance in the free space with a 2.2GHz circuit consisting of a transmitting antenna with a gain of 20dB and a receiving antenna with a gain of 30dB. And the transmitted power is 300W.

Solution:
Given-
Transmitting antenna gain, G_t = 20dB
Receiving antenna gain, G_r = 30dB
Transmitted power, P_t = 300W
Distance, d = 2km.
The maximum power received will be-

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d}\right)^2 = 300 \times 10^2 \times 10^3 \left(\frac{3 \times 10^8}{2.2 \times 10^9 \times \pi \times 2 \times 10^3}\right)^2$$

$P_r = 3 \times 10^7 \times 4.71 \times 10^{-10}$

$P_r = 0.0141W = 14.1$ mW

Rack your Brain

The number of sources required in a binomial array of 11 elements is_____.

Example 21:
An antenna receives 2.5μW power from a radio station. The effective area (in m^2) if the antenna is located in the far zone of the station where E is 60m V/m will be-
(A) 0.5241 (B) 0.623
(C) 0.231 (D) 0.713

Solution:
Given-
Power received, P_r = 2.5μW
Electric field = 60m V/m
Input power density will be-

$$P_{in} = \frac{E^2}{2\eta} = \frac{(60 \times 10^{-3})^2}{2 \times 120\pi}$$

$$P_{in} = \frac{0.0036}{240\pi} = 4.77 \times 10^{-6} W$$

Effective aperture area is given by-

$$A_e = \frac{P_r}{(P_{avg})_{in}} = \frac{2.5 \times 10^{-6}}{4.77 \times 10^{-6}}$$

$$A_e = 0.5241 \ m^2$$

Rack your Brain

The R_{rad} of an antenna radiating 10 kW of power and is fed at 10 amps is _____.

Example 22:
A transmitting antenna with a 700MHz of carrier frequency produces 78W of power. The power received by another antenna at a free space of 1.5km will be-
(A) 0.0403 μW
(B) 0.403 μW
(C) 4.032 μW
(D) 0.0403 mW
[Let both the antennas has unity gain]

Solution:
Given-
Frequency, f = 700MHz = 700×10^6Hz
Transmitting power = 78W
Distance, d = 1.5km
Gain of transmitting antenna = Gain of receiving antenna i.e., $G_t = G_r = 1$
∴ The power received is given by-

$$P_r = P_t \, G_t \, G_r \left(\frac{\lambda}{4\pi d}\right)^2$$

$$P_r = 78 \times 1 \left(\frac{3 \times 10^8}{700 \times 10^6 \times 4\pi \times 1.5 \times 10^3}\right)^2$$

$$P_r = 4.032 \times 10^{-8} = 0.0403 \mu W$$

Rack your Brain

What will be the directivity of a broadside array of 10λ length?

Example 23:
The two lossless X-band horn antennas are separated by a distance of 180λ. The amplitude reflection coefficients at the terminals of the transmitting and receiving antenna are 0.18 and 0.20 respectively. The maximum

directivities are 20dB and 25dB of transmitting and receiving antenna respectively (Over an isotropic antenna). Consider that the input power in the lossless transmission time connected to the antenna is 2.4W and the antennas are perfectly aligned and matched polarized. Calculate the power (in mW) delivered to the load at the receiver-

(A) 15 (B) 2.98

(C) 13 (D) 13.7

Solution:

Given that-

The reflection coefficient of transmitter antenna (Γ_t) = 0.18

The reflection coefficient of receiving antenna (Γ_r) = 0.20.

Here, given that antennas are lossless $R_L = 0$

Maximum directive gain of transmitter antenna (G_t) = 20dB & G_r = 25dB

Distance, d = 180λ

$P_{input} = 2.4$ W

The power received will be-

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d}\right)^2 [1 - |\Gamma_t|^2] [1 - |\Gamma_r|^2] \eta_r \eta_t (PLF)$$

here, PLF is 1 for perfectly matched polarized.

and for lossless transmission η_r, $\eta_r = 1$

$$P_r = 2.4 \times 10^2 \times 10^{2.5} \times \left(\frac{\lambda}{4 \times 180\lambda}\right)^2 \left[1 - (0.18)^2\right][1 - (0.2)^2]$$

$$P_r = 240 \times 10^{2.5} \times 1.95 \times 10^{-7} \times (0.9676)\ (0.96)$$

$\therefore P_r = 13.7$ mW

Example 24:

The electric field of a uniform plane wave traveling along the negative z-direction is expressed as-

$$\vec{E}_i = E_0\, e^{-jkz} \left(\hat{a}_x + j\hat{a}_y\right)$$

This wave is incident upon a receiving antenna placed at the origin and where radiated electric field towards the incident wave is given by-

$$\vec{E}_a = E_l \frac{1}{r} e^{-jkr} \left(\hat{a}_x + 2\hat{a}_y\right)$$

The polarization of the incident wave, the polarization of the antenna and lossless due to the polarization mismatch are, respectively-

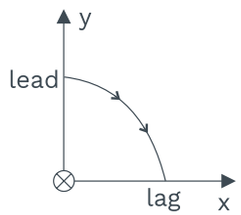
(A) Linear, circular (clockwise), –5dB
(B) Circular (clockwise), linear, –3dB
(C) Linear, circular (clockwise), –5dB
(D) Circular (anti-clockwise), linear, –3dB

Solution:

The electric field of the incident wave is-

$$\vec{E}_i = E_0 e^{j\beta z}(\hat{a}_x + j\hat{a}_y)$$

$\because \quad E_{0x} = E_{0y}$, $\delta = \phi_y - \phi_z = 90° - 0° = 90°$



$$\hat{a}_k = -\hat{a}_z$$

Since the amplitude is the same ($E_{0x} = E_{0y}$)

Therefore, it is circular.

And due to wave traveling in the negative z-direction. By right-hand rule, we can conclude the rotation is clockwise rotation.

Hence, it is observed that polarization is circular and is rotating clockwise.

Also, for polarization of antenna, the electric field is given as-

$$\vec{E}_a = E_l \frac{1}{r} e^{-jkr} (\hat{a}_x + 2\hat{a}_y)$$

here $E_{0x} \neq E_{0y}$ (Amplitude)

Phase difference $\delta = \phi_y - \phi_x = 0$

$\delta = n\pi$

Rack your Brain

Directivity of an end-fire antenna array of 10λ length is_____.

Gray Matter Alert!!!

When the charges around the antenna are set in motion, the other charges are separated from the antenna, and they are also set in motion. The electric and magnetic fields so produced are perpendicular to each other.

$\therefore$ It is linearly polarized.

Now, the losses due to polarization mismatch is given by-

$$PLF = |\hat{\rho}_w \cdot \hat{\rho}_a|^2$$

As we know that-

$$\text{Unit vector} = \frac{\text{Vector}}{\text{Magnitude}}$$

$$\hat{\rho}_w = \frac{\vec{E}_i}{|\vec{E}_i|} = \frac{(\hat{a}_x + j\hat{a}_y)E_0 e^{-j\beta z}}{E_0 e^{-jkz}\sqrt{1^2 + 1^2}}$$

$$\hat{\rho}_w = \left(\frac{\hat{a}_x + j\hat{a}_y}{\sqrt{2}}\right)$$

$$\hat{\rho}_a = \frac{\vec{E}_a}{|\vec{E}_a|} - - = \frac{E_l \frac{1}{r} e^{-jkr}(\hat{a}_x + 2\hat{a}_y)}{E_l \frac{1}{r} e^{-jkr}\sqrt{1^2 + 2^2}}$$

$$\hat{\rho}_a = \frac{(\hat{a}_x + 2\hat{a}_y)}{\sqrt{5}}$$

$\therefore$ $PLF = |\hat{\rho}_w \cdot \hat{\rho}_a|^2$

$$PLF = \left|\left(\frac{\hat{a}_x + j\hat{a}_y}{\sqrt{2}}\right) \cdot \left(\frac{\hat{a}_x + 2\hat{a}_y}{\sqrt{5}}\right)\right|^2 = \left|\frac{1}{\sqrt{10}} + \frac{2j}{\sqrt{10}}\right|^2 = \left|\frac{5}{10}\right| = \frac{1}{2}$$

$\therefore$ PLF (dB) = 10log (1/2) = –3dB

Hence, the correct option is (B).

Example 25:

If an array of 12 isotropic elements is placed along the z-axis at distance $\frac{\lambda}{4}$.

Assuming inform distribution, for an end-fire array. Calculate the following:

(i) Progressive phase shift (radians)

(ii) Half power beamwidth (degree)

(iii) Beamwidth between first null (degree)

(iv) Directivity (dB).

Gray Matter Alert!!!

The gain of a single antenna element is not sufficient in most applications. Hence, the use of antenna arrays fulfills the requirements.

Solution:

(i) The progressive phase shift (λ) is expressed as–

$$\alpha = \pm\beta d = \frac{\pm 2\pi}{\pi} \times \frac{\lambda}{4} = \frac{\pm \pi}{2} \text{ radians}$$

(ii) The half-power beamwidth is expressed as-

$$HPBW = 2\sqrt{\frac{0.88\lambda}{Nd}}$$

$$HPBW = 2\sqrt{\frac{0.88\lambda \times 4}{12 \times \lambda}}$$

HPBW = 1.08 radian or 62.06°

(iii) The beamwidth between first nulls is given by-

$$BWFN = 2\sqrt{\frac{2\lambda}{Nd}}$$

$$BWFN = 2\sqrt{\frac{2 \times \lambda \times 4}{12 \times \lambda}} = 2\sqrt{\frac{8}{12}}$$

BWFN = 1.632 or 93.56°

Rack your Brain

B.W.F.N. of broadside array of length 10λ is_____.

(iv) Directivity will be–

$$D = \frac{4Nd}{\lambda} = \frac{4 \times 12 \times \lambda}{4 \times \lambda}$$

D = 12

In decibels (dB)-

$$(D)_{dB} = 10 \log 12 = 10.79 \text{ dB}$$

Example 26:

The broadside array of 18 isotropic antenna elements which are equally separated by $\frac{\lambda}{2}$ distance and then the beamwidth between the first null will be–

(A) 11.46°
(B) 12.60°
(C) 12.13°
(D) 13.16°

Solution:
For broadside-array, the beamwidth between the first null is expressed as-

$$BWFN = \frac{2\pi}{Nd}$$

Here, N = no. of elements = 18 and $d = \frac{\lambda}{2}$

$$\therefore BWFN = \frac{2\lambda \times 2}{18 \times \lambda} = \frac{4\lambda}{18\lambda} \text{radian}$$

BWFN = 0.22 rad or 12.60°
Hence, the correct option is (B).

Rack your Brain

HPBW of the end-fire antenna array of length 10% is 0.894

Example 27:
A lossless antenna with an 8 dB of directive gain is considered. If 2mW of power is fed, then the total power radiated will be–
(A) 12.61 mW (B) 13 mW
(C) 12 mW (D) 14.18 mW

Solution:
Given that D = 8 dB &
Power input (P_{input}) = 2mW = 2×10^{-3} W
As we know that-
Power Gain (G_P) is expressed as-

$$G_P = \left(1 - |\Gamma|^2\right) \eta \times D$$

Given, that the antenna is lossless.

i.e., $R_l = 0$ and $\eta = 1$

Assuming matched condition i.e., reflection is negligible.
or $\Gamma = 0$
And $G_P = D$

$$\text{or } D = \frac{P_{rad}}{P_{input}}$$

$$P_{rad} = D \times P_{input}$$

$P_{rad} = 10^{0.8} \times 2 \times 10^{-3} = 12.61$ mW

Hence, the correct option is (A).

Gray Matter Alert!!!

Electromagnetic waves carry information signals from the transmitter to the receiver. There is no communication system without only one type of antenna or the other.

Example 28:
A transmission line is feeding 1.5W of power to a horn antenna. The antenna is matched to a transmission line and the gain of the horn antenna is 8 dB. The total power radiated by the horn antenna into the free space will be–

(A) 10 W
(B) 9.46 W
(C) 9 W
(D) 8.46 W

Solution:
Given that –
G_p = 8 dB
Power input = 1.5W
Given that the antenna is matched.
Thus, there will be no reflections ($\Gamma = 0$)
and assuming lossless antenna,

$R_l = 0$

$\therefore \eta = 1$

Thus, the Power gain = Directivity
and Radiated power (P_{rad}) will be-

$P_{rad} = P_{in} \times G_P$

$P_{rad} = 1.5 \times 10^{0.8} = 9.46$ W

Hence, the correct option is (B).

Rack your Brain

The number of secondary lobes mainly depends on the number of nulls in the resultant pattern. (True/False)

Rack your Brain

Antenna wires are associated with electric and magnetic fields?

Example 29:
Consider a uniform linear array consisting of 4 isotropic radiating elements separated $\frac{\lambda}{6}$ distance apart.
The progressive phase shift between the elements for forming the main beam at 60° for the end-fire array is–

(A) $\frac{\pi}{6}$ rad
(B) $\frac{-\pi}{6}$ rad
(C) $\frac{-\pi}{4}$ rad
(D) $\frac{\pi}{6}$ rad

Solution:
Given–
Number of elements, N=4
Distance between elements, $d = \frac{\lambda}{6}$
and θ = 60°

As the mean beam is forming then there will be maximum radiation.
Thus, $\psi = 0$

$\therefore \alpha + \beta d \cos\theta = 0$

$\alpha = -\beta d \cos 60°$

$\alpha = -\frac{2\pi}{\lambda} \times \frac{\lambda}{6} \times \cos 60°$

$\alpha = \frac{-\pi}{3} \cos 60° = \frac{-\pi}{6} \text{rad}$

Rack your Brain

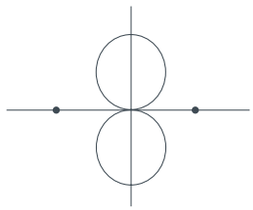
The top-loading is sometimes used with an antenna to increase its bandwidth.

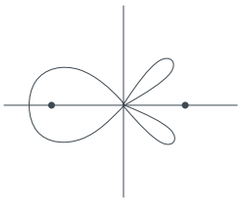
Hence, the correct option is (B).

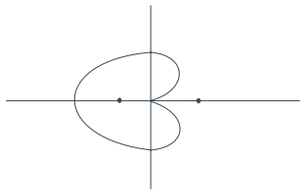
Example 30:

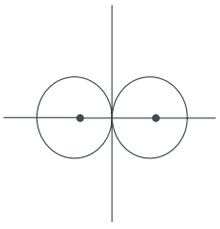
The two identical and parallel dipole antennas are spaced by $\frac{\lambda}{4}$ distance in the H-plane.

They are fed with equal currents but the right most antenna has a phase shift of +90°. The radiation pattern is given as–

(A) 

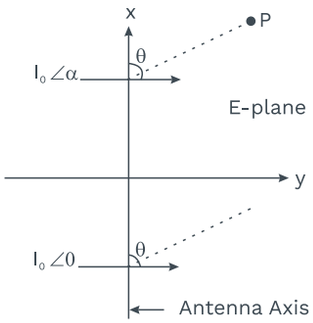
(B) 

(C) 

(D) 

Solution:

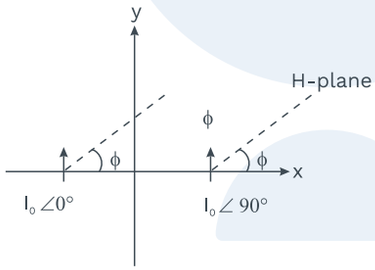
As we have studied earlier that if antenna array elements are placed horizontally at the z-axis and parallel to the y-axis.



The progressive phase shift from the rightmost antenna or topmost antenna to the next antenna.

$\therefore \alpha - 0 = \alpha$

Now, according to the given question, the plane is H-plane.



Given that the phase shift of the right most antenna is 90°.

The progressive phase shift = 90°-0°

$\therefore \alpha = 90°$

The array factor (AF) will be -

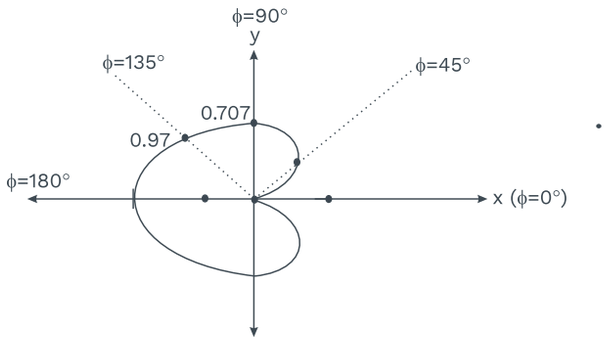
$$|AF|_{norm} = \cos\frac{\psi}{2} = \cos\left(\frac{\alpha + \beta d\cos\phi}{2}\right)$$

$$|AF|_{norm} = \cos\left(\frac{\frac{\pi}{2} + \frac{2\pi}{\lambda} \times \frac{\lambda}{4}\cos\phi}{2}\right)$$

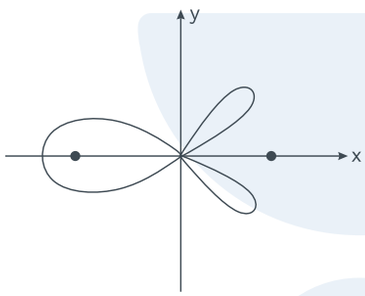
$$|AF|_{norm} = \cos\left(\frac{\pi}{4} + \frac{\pi}{4}\cos\phi\right)$$

Rack your Brain

In a dipole antenna, the two ends are at equal _____with respect to the mid-point.



After pattern multiplication,
The radiation pattern will be –



Hence, the correct option is (B).

PRACTICAL ANTENNAS

- As we know that the half-wave dipole and other has a radiation pattern distributed in all directions (isotropic) or in the form of a donut.
- But we don't always require this complete horizontal pattern distributed/ covered in all directions.
- Sometimes we required the radiation pattern in a particular direction and this focused approach of the radiation pattern is achieved in Yagi-Uda antenna, Parabolic-reflector antenna, etc.
- The salient features of the Yagi-Uda antenna are–
  (i) It is used as a TV reception antenna. It has the capability of capturing signals in a particular direction only.
  (ii) High focus or directivity in a required particular direction.
  (iii) Yagi-Uda antenna used in folded dipole antenna characteristics.
  (iv) For achieving high directivity, some of the parasitic elements are required elements.
- Yagi–Uda were the two Japanese scientists named Shintaro Uda and his colleague Hidetsugu Yagi who invented and designed this antenna in 1926.

Yagi–Uda antenna is generally coined as the directional antenna which consists of two or more resonant array elements in an end-fire array.

- Earlier when TV was not broadcasted by satellite, in that era the Yagi-Uda antenna was designed and widely used.
  This antenna has to look in the direction of the TV for getting high signal.
- The elements on Yagi-Uda are active elements which are usually metal rods behaving like half-wave dipoles.
- The Yagi-Uda antenna consists of a dipole as the main radiating (or driven element) to which the power is directly applied from the feeder.

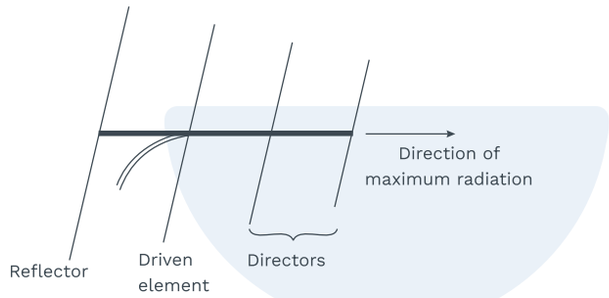


Figure 8.24: Basic concept of Yagi-Uda Antenna

- In other words, Yagi-Uda antennas include a single driven element that is usually connected to a radio transmitter and/or receiver through the transmission line and “parasitic elements” with no electrical connections, including reflectors and any number of directors.
- The active element is used connected with the TV.
  Since, it has radiation patterns in all directions which is not needed.
  Thus, we install reflectors elements in the antenna design.
- Reflector elements were used usually only one and are slightly greater in size from driven dipole and are generally placed behind the driven element
- Reflectors are generally placed in the opposite directions of the required transmission.

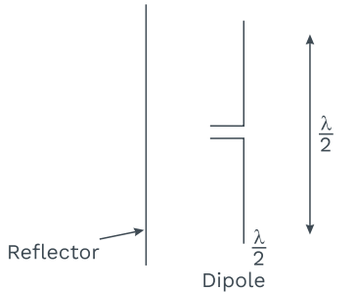


Figure 8.25

- The reflectors are placed near the active/ driven element (λ/2 length). A transmission line, open-circuited and with 5% longer in length and a closed spacing of $\frac{\lambda}{10}$.
- The longer length reflector and $\frac{\lambda}{10}$ spacing is required as it helps in the cancellation of the radiation on the side facing driven element and provides unidirectionality.
- If the half-wave dipole act as a resonant LC circuit, the reflector behaves as an inductive in nature having 90° lead or lag phase shift.
- So, the radiation coming into half-wave dipole will be in oscillatory condition and the radiation coming to reflector will be leading or lagging.
- And this leading or lagging in reflector causes cancellation of both radiation patterns.
- Thus, we get a smaller radiation pattern on the left side of the driven element and a larger radiation pattern on the other half side of the driven element.

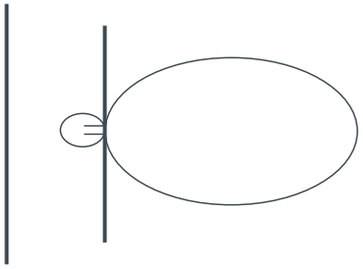


Figure 8.26

Hence, because of the reflector, the half-wave dipole radiates only in one direction.

Rack your Brain

In coaxial cables or waveguides, the transmission loss per unit length is dependent on the distance. (True/ False)

Rack your Brain

In an end-fire array, there exists no radiation at _____ angle to the axis of the array.

- The reflector has a 90° phase shift and the arrays always have a nature of cancellation depending upon the phase angle and spacing.
- Now, to improve efficiency further the directors are added to this antenna design.
- These directors are generally placed on the right side of the driven element and are 5% lesser in length.
- And if the reflectors have 90° phase shift in it, then the director will have –90° phase–shift in it with capacitive nature.

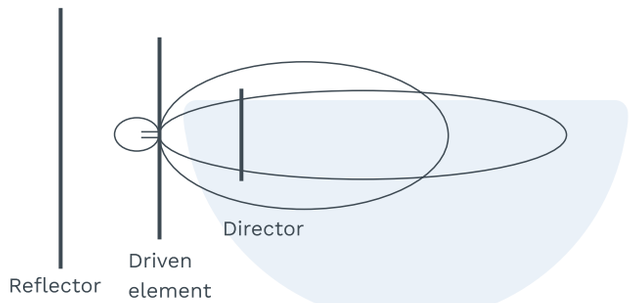


Figure 8.27: Radiation pattern of Yagi-Uda antenna

- The capacitive nature of a director helps in increasing the directivity & adds to the radiation and hence called a director.
- Using multiple directors helps in increasing radiation in the required direction. Since, it enhances the directivity.

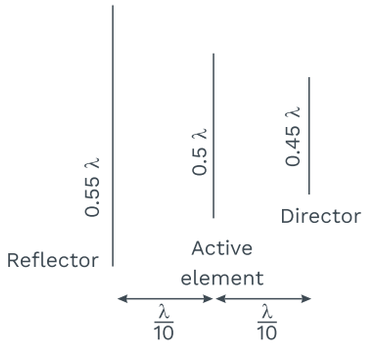


Figure 8.28: Antenna elements

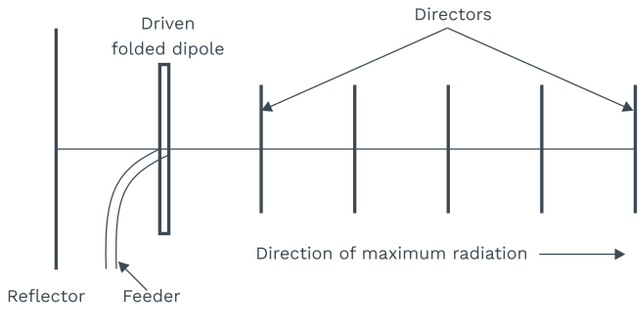


Figure 8.29: Yagi-Uda Antenna

- It is noted that resonant circuit has LC circuits producing oscillations, longer length transmission line.
- It can be noted that resonant circuit has LC-LC oscillations producing radiation, longer line length has inductive has capacitive dominance. Thus, array factor or radiation gets added on one side and cancelled on another side.
  These are referred to as parasitic array and depends on induction fields.
- Reflectors and directors are included in the parasitic elements category. Because they are not connected to the transmission line but are located close enough that induction fields work on them.
- This influence of induction fields allows to cancel the radiation on one side of the driven (active) element and adds on to the other side. Thus, it behaves like a directional antenna.
- Parasitic elements can be related as they persist in parasitic behavior. These elements are not connected to the feed but vanishing the radiation of the active element on one side of the antenna.

Radiation Pattern of Yagi–Uda Antenna

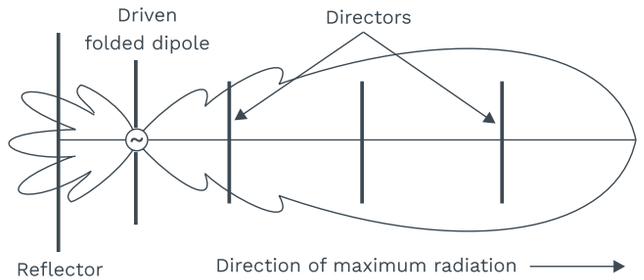


Figure 8.30

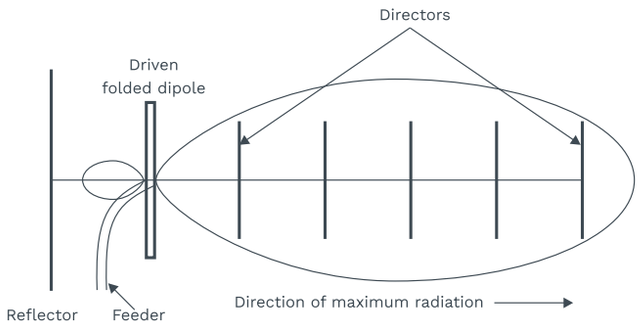


Figure 8.31

- Multiple directors can be used to achieve the narrow beam in radiation pattern or to increase gain.
  So, according to the application of antenna number of directors are installed.

Optical Model of Yagi-Uda Antenna

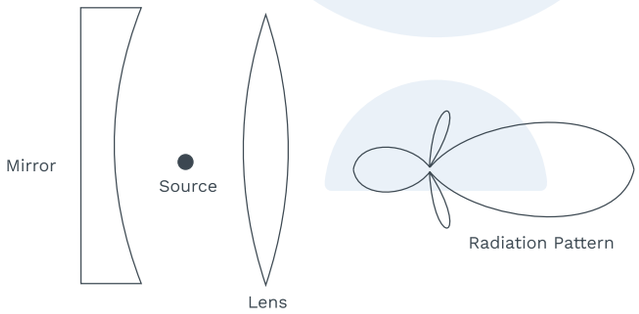


Figure 8.32: Optical Model of Yagi-Uda Antenna

- The optical model helps us to understand the concept of the Yagi-Uda antenna
  Here, the reflector acts as the mirror while the directors act as the lens
  Mirror works the same as reflectors by reflecting back the radiation / or majority canceling the radiation.
  The lens acting like directors helps the radiation to be more focused in a particular direction.
- If we increase the number of parasitic elements (or directors) reduced radiation is caused because the power is mainly wasted in inducing the side-by elements rather than radiating into the sky.
  Hence, radiation gets lost due to the increasing number of parasitic elements.

- If we use 73 Ω and λ/2 dipole the radiation resistance drops to 23 Ω or radiated power gets reduced.
  To overcome this problem, we use folded dipole antenna.

Folded dipole antenna

- A folded antenna consists of two conductors connected on both sides.
- It is folded in such a way to form a cylindrical closed shape.
- The feed is given at the center of the dipole. And the length of the dipole is half of the wavelength.
- The folded dipole antenna has bi-directional directivity. And the input impedance is higher. The frequency range is 3 kHz to 300 GHz and is widely used in television receivers.
- The folded dipole antenna is used to increase the radiation resistance to a moderate level.

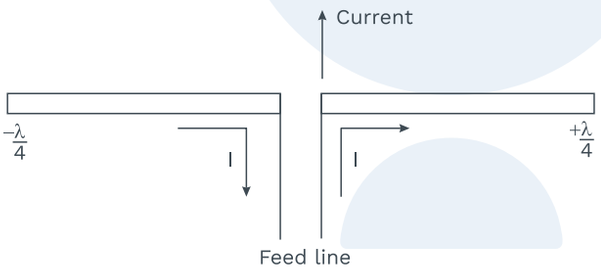


Figure 8.33

- Advantages of Half-wave folded dipole-
  (i) Good reception of the balanced signal
  (ii) This receives a particular signal from a band of frequencies without losing the quality.
  (iii) It maximizes the signal strength.

Microwave Dipole Antenna

- Microwave antennas are used to broadcast microwave transmission between two or more locations
- Microwave antennas can transmit few GHz frequencies
- These antennas have a length in centimeters wavelength.
- The range of operation is thousands of kilometers instead of being in small size.

Antenna Type

Reflector types-Parabolic antenna

- The parabolic antenna uses parabola-shaped reflectors (curved surface with the cross-sectional of a parabola) to direct the radio waves. Generally, the diameter of this antenna ranges from 6" to 12' , depending upon the requirement of the user.

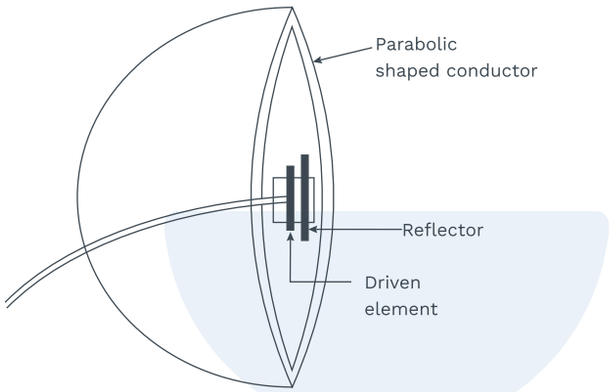


Figure 8.34

- Parabolic dish antenna has a circular mouth and 3D parabolic reflector which is huge with a large capturing area to get more focus.
- On increasing the capturing area, the range of operation is also increased.

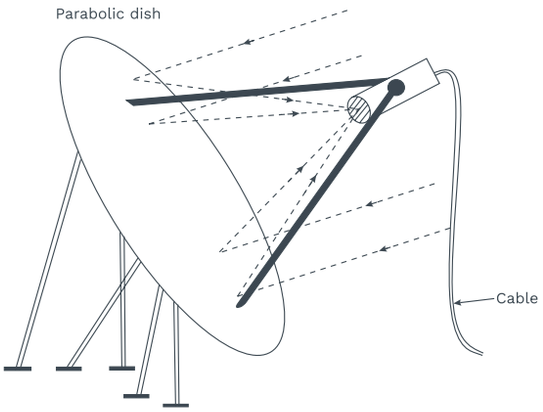


Figure 8.35: Parabolic Dish

Horn antenna

- It consists of a horn-shaped metal waveguide to direct radio waves into a beam.
- A horn antenna may be regarded as a flared-out (or opened-out) waveguide.

- The main function of the horn shape is to provide a uniform phase front with a larger aperture than that of the waveguide and hence greater directivity.
- Jagadish Chandra Bose constructed a pyramidal horn in 1897.
- For UHF and microwave frequencies greater than 300 MHz, horn antennas are generally used.
- The wire-antennas (monopole or dipole) are usually involved in the category of resonant antennas.
- The main reason behind the popularity of horn antenna is its ease of fabrication as it can be fabricated using a section of the rectangular metallic waveguide by simply flaring its outer edges.
- Another reason for wider practical usage of horn antenna is their higher directivity, low loss, and their effective usage at the transmitter side for high-power applications.
- The electromagnetic horns are also practically used for various other applications such as the communication system, the radio astronomy, microwave imaging and non-destructive testing, etc.
- The electromagnetic horn provides a smooth transition between the rectangular waveguide supporting the dominant TE_{10} mode and the free space impedance by flaring the waveguide.
- The horn utilizing rectangular waveguide geometry are classified into three categories namely, the E-plane sectoral horn, the H-plane sectoral horn, and the pyramidal horn antenna (as shown in the figure).

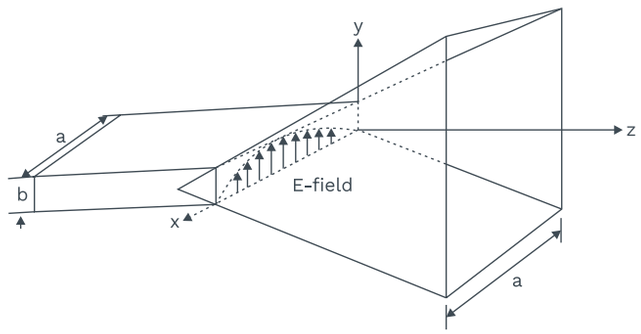


Figure 8.36(a): Rectangular horn antenna (E-plane)

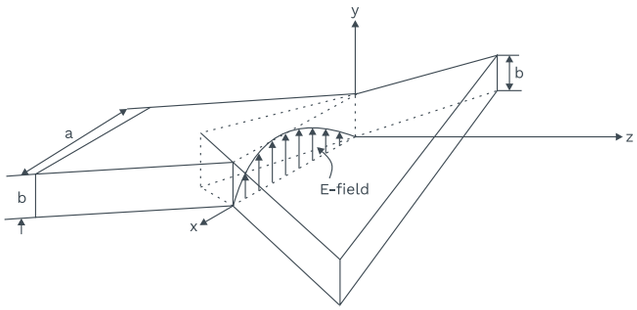


Figure 8.36(b): Rectangular horn antenna (H-plane)

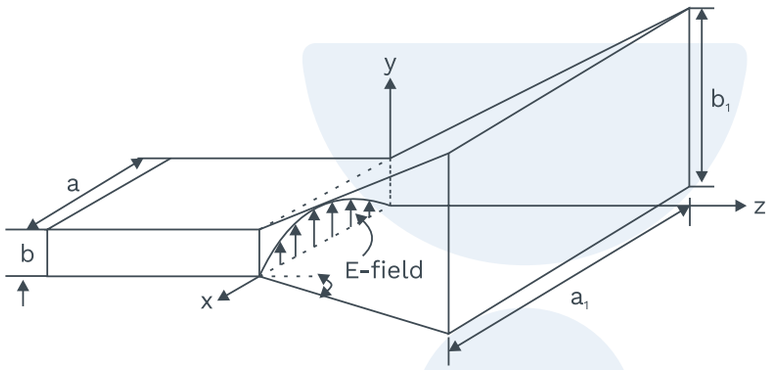


Figure 8.36(c): Pyramidal horn antenna

- It is to be noted that for the rectangular waveguide operating in the TE_{10} mode, the E-plane contains the vertical narrow wall because the maximum electric field is oriented along the vertical y-axis.
- Similarly, the H-plane for the feeding rectangular waveguide in the present situation contains the horizontal broader wall due to the orientation of the maximum magnetic field in this plane.
- If the horn is flared along the E-plane, keeping the broader dimension of the feeding rectangular waveguide unchanged, then it is known as the E-plane sectoral horn antenna as shown in figure 8.37 (a).

Rack your Brain

The antenna radiates maximum when it is in perfect resonant condition. (True/False)

- For the H-plane sectoral horn antenna depicted in Figure 8.37 (b), the flaring is done along the H-plane, keeping the dimension b of the rectangular waveguide constant. In the case of the pyramidal horn, the flaring is done along both the planes as illustrated in Figure 8.37 (c), and hence its characteristics are the combination of E-plane and H-plane sectoral horn antennas.
- Out of these three horn structures, the pyramidal horn antenna is the most popular one.

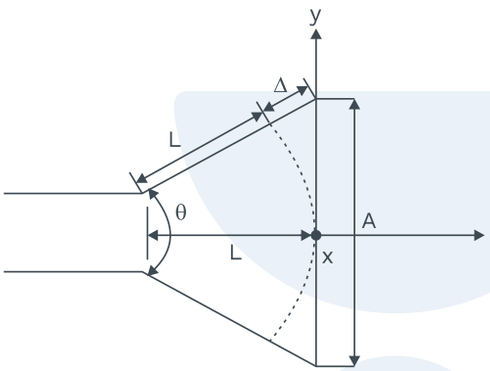


Figure 8.37: Pyramidal horn antenna

- The cross-section of a typical pyramidal horn antenna, which may be applied to both the E-plane and the H-plane by changing the flare angle and the aperture dimension 'A' , is shown in Figure 8.37.
- From this figure, it may be clearly observed that the phase at the edge of the horn aperture will not be the same that at the center of the horn aperture. This is because the wave travels different distances along the side and at the center axis of the horn, thus creating the path difference 'A' as shown in this figure. The geometrical cross-section provided in the Figure 8.37 may be approximated by two triangles at the upper half and the lower half, thus providing the following simplified relationship-

Rack your Brain

Where an electric field is developed at the ends of an antenna?

$$\cos\frac{\theta}{2} = \frac{L}{L+\Delta}$$

$$\tan\frac{\theta}{2} = \frac{A}{2L}$$

where-

θ = Flare angle (θ_e for E-plane and θ_h for H-plane)

A = Aperture dimension ('b' for E-plane and 'a' for H-plane).

L = Horn length

- For the practical realization of the horn antenna, 'A' is usually chosen to be equal to 0.25λ or less in the E-plane. However, the value of 'A' is usually taken to be larger in the H-plane with its approximate value given by 0.4 λ.
- It is noted that the gain of the horn antenna can be enhanced by increasing the aperture dimension 'A' for a pyramidal horn of length 'L'.

Rack your Brain

The grounded vertical antenna of length $\lambda/4$ is an antenna that radiates equally in all directions. (True/False)

CHAPTER SUMMARY

- Radiation field equations are given by-

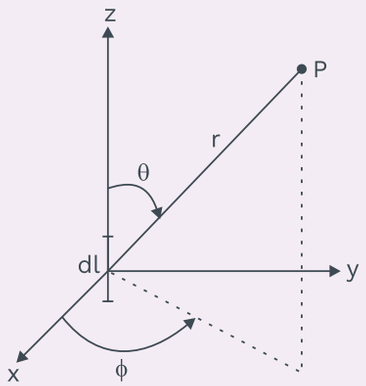
$$H_{\phi s} = \frac{I_0 dl}{4\pi}\sin\theta\left[\frac{j\beta}{r} + \frac{1}{r^2}\right]e^{-j\beta r}$$

$$H_{rs} = 0 = H_{\theta s}$$

$$E_{rs} = \frac{\eta I_0 dl}{2\pi}\cos\theta\left[\frac{1}{r^2} - \frac{j}{\beta r^3}\right]e^{-j\beta r}$$

$$E_{\theta s} = \frac{\eta I_0 dl}{4\pi}\sin\theta\left[\frac{j\beta}{r} + \frac{1}{r^2} - \frac{j}{\beta r^3}\right]e^{-j\beta r}$$

$$E_{\phi s} = 0$$



1. $\frac{1}{r}$ term is called the far-field or radiation field because it is the only term that remains at the far zone.
2. $\frac{1}{r^2}$ term is called the induction field.
3. $\frac{1}{r^3}$ term is called the electrostatic field.

- At far field $r >> \frac{1}{\beta}$

$$H_{\phi r} = \frac{jI_0\beta dl}{4\pi r}\sin\theta e^{-j\beta r},\ E_{\theta r} = \eta H_{\phi r}$$

$$H_{rs} = H_{\theta s} = E_{rs} = E_{\phi s} = 0$$

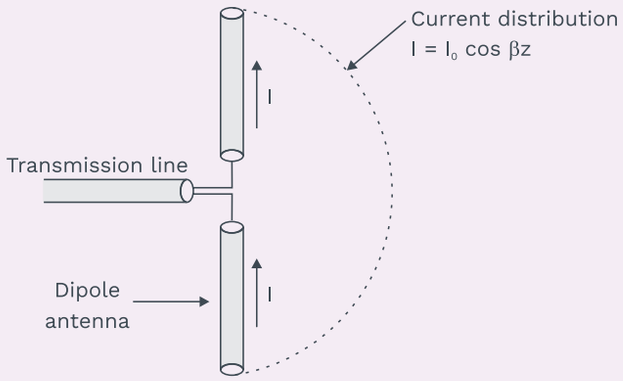
- Radiation resistance is given by -

$$R_{rad} = 80\pi^2 \left(\frac{dl}{\lambda}\right)^2$$

- The distance at which induction field and radiation field have equal amplitude is -

$$r = \frac{\lambda}{2\pi}$$

Half wave dipole antenna (l = λ / 2)



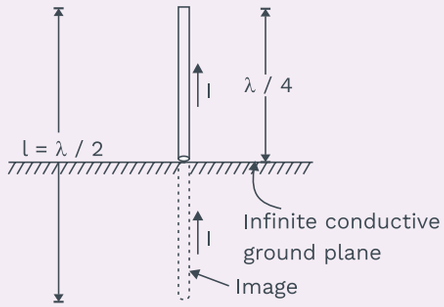
- It is a center-fed antenna and horizontally installed so that the electric field vector is always parallel to the surface of the earth.
- Non-uniform current distribution along its length and hence effective length depends upon the type of current distribution.
- At the far-field,

$$H_{\phi s} = \frac{jI_0 e^{-j\beta r} \cos\left(\frac{\pi}{2}\cos\theta\right)}{2\pi r \sin\theta}, \quad E_{\theta s} = \eta H_{\phi s}$$

- Radiation resistance is approximate-

$$R_{rad} = 73\ \Omega$$

Quarter-wave monopole antenna



- It consists of half of a half-wave dipole antenna located on a conducting ground plane.
- It is a vertically polarized antenna so that the electric field vector is always perpendicular to the surface of the earth.
- It is end fed antenna.
- Using image theory, we replace the infinite perfectly conducting plane with the image of the monopole, hence it produces the same far-field that of half-wave dipole (Above the ground).
- Radiation resistance is given by -

  $R_{rad} = 36.5\ \Omega$
- At far-field,

$$H_{\phi s} = \frac{jI_0 e^{-j\beta r}\cos\left(\frac{\pi}{2}\cos\theta\right)}{2\pi r\sin\theta},\ 0 \le \theta \le \frac{\pi}{2}$$

$$H_{rs} = H_{\phi s} = E_{rs} = E_{\theta s} = 0$$

Antenna Parameters:

S.No.	Parameters	Definition	Formula
1.	Antenna impedance	It is defined as the ratio of input voltage to the input current.	$Z_a = \frac{V_i}{I_i} = R_a + jX_a$ $R_a = R_r + R_L$ R_r = Radiation Resistance R_L = Loss radiation
2.	Radiation resistance	It is defined as fictitious or hypothetical resistance that would dissipate an amount of power equal to radiated power.	$R_{rad} = \frac{P_{rad}}{I^2_{rms}}$
3.	Radiation intensity	It is the strength of the power of a radiated electromagnetic wave in any direction from the antenna.	$U(\theta,\phi) = \frac{\text{Radiated Power}}{\text{direction}}$ $U(\theta,\phi) = \frac{P_{rad}}{d\Omega}$ W / Sr
4.	Directive gain	Directive gain is defined as the ratio of radiation intensity in that direction to the average radiation intensity.	$G_d(\theta,\phi) = \frac{U(\theta,\phi)}{U_{avg}(\theta,\phi)}$ $G_d = \frac{4\pi U(\theta,\phi)}{\int_{\theta=0}^{\pi}\int_{\phi=0}^{2\pi} U(\theta,\phi)\sin\theta d\theta d\phi}$
5.	Directivity	The directivity of the antenna is the ratio of maximum radiation intensity to average radiation intensity.	$D = \frac{U(\theta,\phi)_{max}}{U_{avg}(\theta,\phi)} = G_{dmax}$

S.No.	Parameters	Definition	Formula
6.	Power gain	Power gain is the ratio of radiation power intensity to the input power.	$G_P(\theta,\phi) = \frac{4\pi U(\theta,\phi)}{P_{in}}$
7.	Front to back ratio (FBR)	FBR is defined as the ratio of radiated power in the desired direction to the radiated power in the opposite direction.	$FBR = \frac{P_r \text{ in desired direction}}{P_r \text{ in opposite direction}}$
8.	Antenna beamwidth	Antenna beamwidth is the measure of the directivity of the antenna.	$D = \frac{4\pi}{\Omega_A} = \frac{4\pi}{B}$ where- D = directivity Ω_A = Solid angle B = Beam area

- An antenna array is a system of similar antennas oriented serially to get greater directivity in the desired direction.
- An antenna array is a group of radiating elements arranged to produce particular radiation characteristics.
- In a uniform linear array, the elements are fed with a current of equal magnitude & having a uniform progressive phase shift along the line.
- Types of antenna array:
  (a) Broadside array (b) End fire array
- Array factor is the value of normalized electric field intensity.
  It is defined as

$$AF = \frac{|E_T|}{|E_T|_{max}}$$

- The antenna array factor of the two-point sources is given by :

$$AF = \cos\left[\frac{\beta d \cos\theta + \alpha}{2}\right]$$

S.No.	Parameters	Broadside array	End fire array
1.	Definition	An arrangement in which the principal direction radiation is perpendicular to the array axis and also to the plane containing the array element.	An arrangement in which the principal direction of radiation coincides with the direction of the array axis.
2.	Maximum radiation occurs	Perpendicular to the array axis i.e., at 90°	Along the axis of the array i.e., at 0°
3.	Progressive phase shift	$\alpha = 0°$	$\alpha = \pm\beta d$

Pattern multiplication

- Pattern multiplication describes the resultant field pattern.
- The pattern multiplication rule only applies to an array consisting of identical elements.
- Resultant pattern = Unit pattern × Array pattern (also called as group pattern)

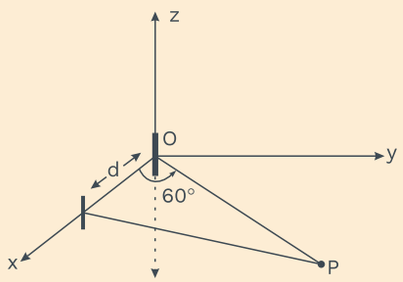
$$= (\sin\theta \text{ or } \cos\theta) \times \cos\left(\frac{\beta d \cos\theta + \alpha}{2}\right)_{\text{E–plane}}$$

- Radiation pattern due to array factor is known as a group pattern.
- Pattern multiplication is useful for designing an array.

PRACTICE QUESTIONS

Q1 **If the BWFN of a uniform linear array of N equally distanced (d) elements which are equally excited antenna elements is determined by-**
(A) N only but not on d.
(B) d only but not N.
(C) the ratio of N and d.
(D) Product of N and d.

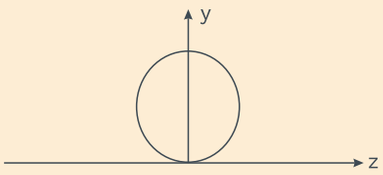
Q2 **If the two half-wave dipole antennas are shown in the figure. The antennas are excited by sinusoidal varying currents of frequency 4 MHz; with a phase shift of $\frac{\pi}{2}$ between then, the element at the origin and E-field at point P in the x-y plane occurs at the 60°.**



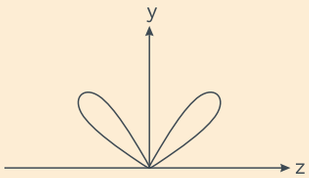
(A) 37.5 m
(B) 38.5 m
(C) 36.5 m
(D) 34.5 m

Q3 **A $\frac{\pi}{2}$ dipole is kept horizontally at a height $\frac{d}{2}$ above a perfect conducting infinite around the plane. The radiation pattern in the plane of the dipole (E-plane) looks approximately as**

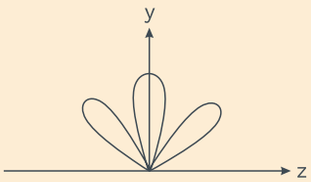
(A)



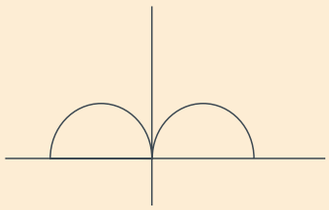
(B)



(C)



(D)



Q4 **The resultant pattern of an array of two-directional but point sources short collinear dipoles.**

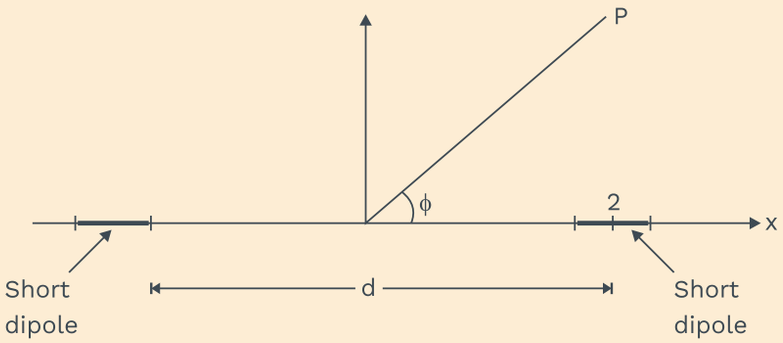
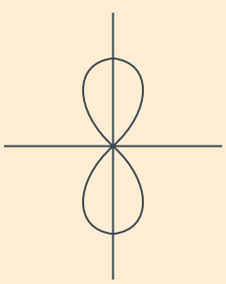
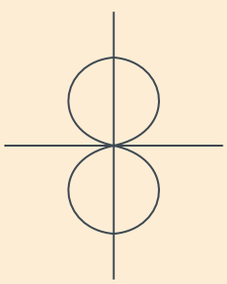


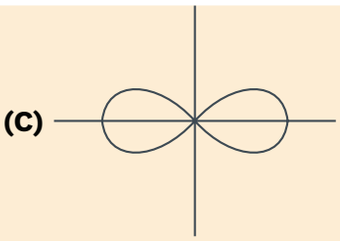
Figure: An array of short collinear dipoles.

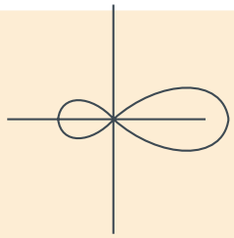
(A)



(B)



(C) 

(D) 

Q5 **Determine the gain of a two-element broadside array having a spacing of λ/2 distance between them.**

(A) 2 **(B) 0.5** **(C) 4** **(D) $\frac{1}{4}$**

Q6 **The power radiated by a quarter wave dipole having a current of I_{rms} will be-**

(A) I^2 0.609η/π
(B) I^2 0.609η/2π
(C) I^2 0.609π/η
(D) I^2 0.609π/2η

Q7 **Which of the following antennas can be considered to have a uniform current flowing along its length?**

(A) Antenna with length 3m at 50MHz
(B) Antenna with a radiation resistance of 30Ω
(C) Wire antenna with a radiation resistance of 1Ω
(D) A perfectly resonant wire antenna

Q8 **The ratio of the effective length to the physical length will be minimum in the case of-**

(A) Half-wave dipoles.
(B) Hertzian dipoles.
(C) Vertically grounded Marconi antenna.
(D) Antenna with the linear current along its length.

Q9 The ratio of effective length to the wavelength will be smallest for-
(A) Hertzian dipole.
(B) Half-wave dipole.
(C) Vertically grounded quarter-wave monopole antenna.
(D) Antenna with the linear current along its length.

Q10 An antenna has a directive gain of 6dB and it is radiating 200mW into a receiving system directed towards the principal maximal of this antenna. What is the power induced in the same receiving system if the antenna is replaced by an isotropic antenna with the same input power?
(A) 50mW (B) 60mW (C) 45mW (D) 50W

Q11 The electric-field plot (vertical pattern) of a wire antenna is found to be majority focused on the direction of from 30° to 90° with its axis. Consider the horizontal pattern is isotropic. Then the antenna's gain will be ____.
(A) 1.91 (B) 4.1 (C) 2.32 (D) 0.91

Q12 The beam angle of a microwave satellite high gain parabolic antenna depends on which of the following options?
(A) Ratio of frequency to diameter mouth.
(B) Product of frequency and diameter of the mouth.
(C) Inverse to the product of frequency and diameter of the mouth.
(D) Diameter to the frequency.

Q13 The directive gain function of wire antenna depends on θ (angle with dipole axis) when it is vertically placed on conducting earth and given as $5\sin 2\theta$ with $0 < \theta < \frac{\pi}{2}$. Determine the received electric field strength at a 5km distance in the direction of half-power point if transmitted power is 1kw.
(A) 30 (B) 3 (C) 0.030 (D) 1.5

Q14 Determine the beam solid angle of spacing distance and currents has a phase shift of 90°.

(A) π rad (B) $\frac{3}{2}\pi$ rad (C) 2π rad (D) $\frac{\pi}{2}$ rad

Q15 If the two dipoles of gain 1.64 each are used for transmitting and receiving antenna purposes. The distance between these two dipoles is 12 m and the radiated power by the transmitting antenna is 20 W at the frequency of 50 MHz.
Evaluate the receiving power.

(A) 40.12 mW (B) 63.87 mW (C) 30.13 mW (D) 85.16 mW

Q16 Calculate the Null-to-Null beamwidth of an end-fire array when array length is 8λ and N = 20.

(A) 0.232 radian (B) 0.081 radian
(C) 0.223 radian (D) 0.314 radian

Q17 An array of isotropic radiators is operated at a frequency of 5 GHz and is required to produce a broadside beam.
The beam width between first nulls for the array length of 12 m is given as-

(A) 1 radian, 4 (B) 0.01 radian, 400
(C) 400 radian, 0.01 (D) 4 radian, 0.04

Q18 If a uniform linear array is required to produce an end-fire beam when it is operated at a frequency of 8 GHz. It contains 40 radiators and are placed at 0.75λ distance apart. The progressive phase shift required to produce the end-fire beam and the array length will be -

(A) 1.5 π rad, 1.125 m (B) 0.5 π rad, 1.6 m
(C) 0.5 π rad, 1.125 m (D) 1.5 π rad, 2.175 m

Q19 An antenna whose radiation resistance is 40 Ω operates at a frequency of 2 GHz with 3.5 A of current flowing in it. The power radiated is _____ W.

(A) 4900 (B) 4000 (C) 4131 (D) 5000

Q20 A circular loop antenna has a 1.25 of diameter. The directivity and radiation resistance are -

(A) 2.656, 2325 Ω (B) 3.131, 2000 Ω

(C) 2.31, 2131 Ω (D) 2.05, 3121 Ω

Answer Key:

1.	(D)	2.	(A)	3.	(B)	4.	(A)	5.	(A)
6.	(B)	7.	(C)	8.	(D)	9.	(A)	10.	(A)
11.	(C)	12.	(C)	13.	(C)	14.	(C)	15.	(D)
16.	(C)	17.	(B)	18.	(A)	19.	(A)	20.	(A)